CG24
Rendering Competition

WEI WANG, JIE-YING LEE

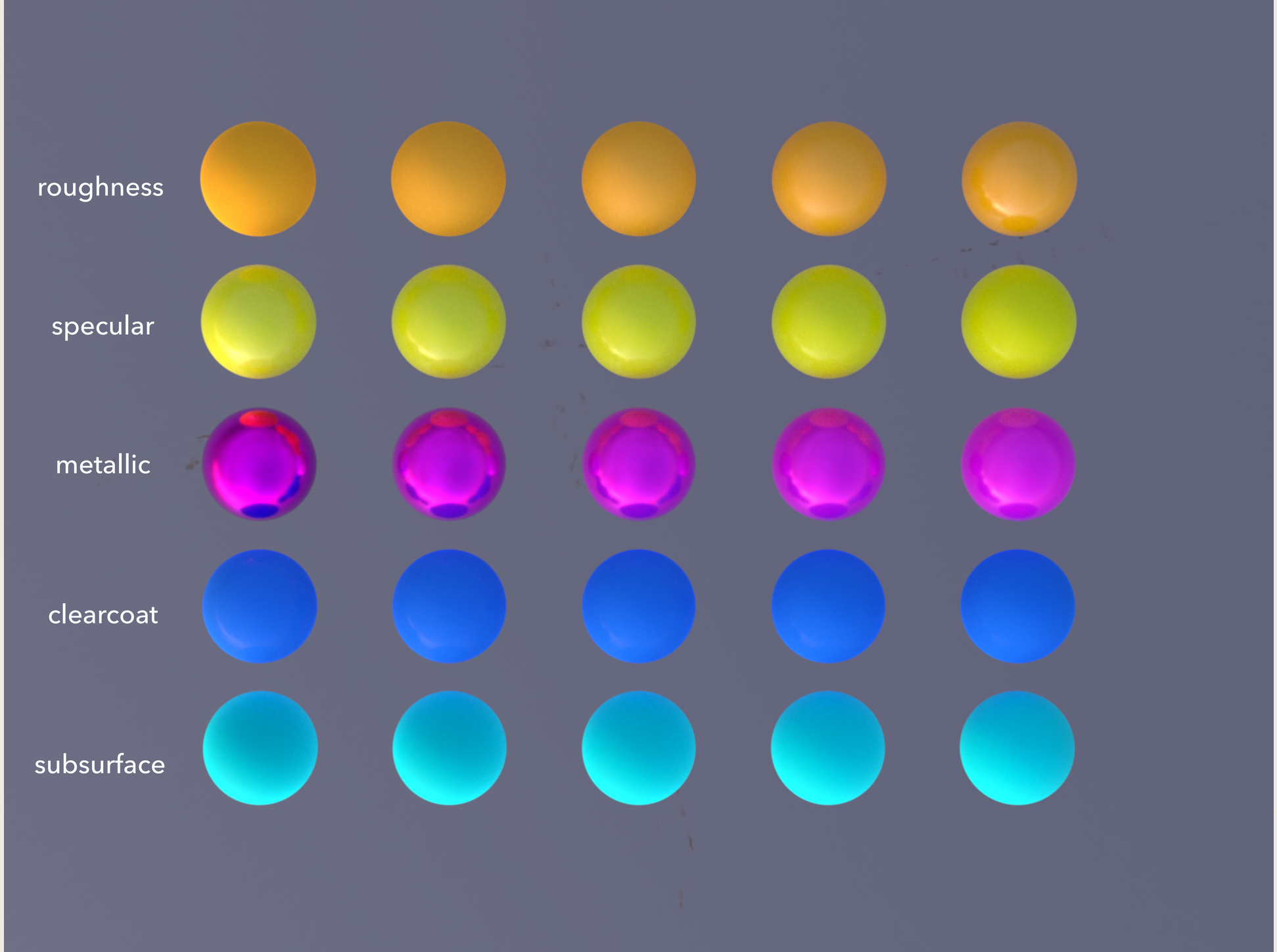
Overview

- Images as Textures
- Procedural Textures - Perlin noise
- Intel’s Open Image Denoise
- Rendering using Euler Cluster
- Stratified Sampling
- **Disney BSDF**
- Realistic Camera Model
- Simple Extra Emitters
- Anisotropic Phase Function
- Extra Reconstruction Filter
- Modeling Meshes
- Equiangular Sampling of Single Scattering
- **Heterogeneous Participating Media**
- Environment Map Emitter

Disney BSDF

Disney BSDF - Importance Sampling

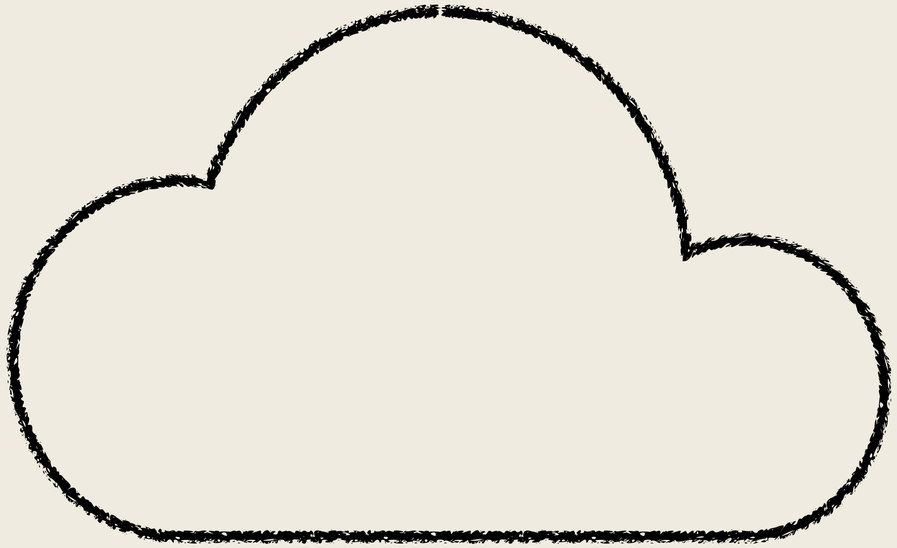
	diffuse	primary specular (specular reflection)	secondary specular (clearcoat)
weight	1.0 - metallic	1.0	0.25 * clearcoat
sampling	cosine weighted hemisphere	isotropic GTR2	isotropic GTR1


roughness
specular
metallic
clearcoat
subsurface

Heterogenous Participating Media

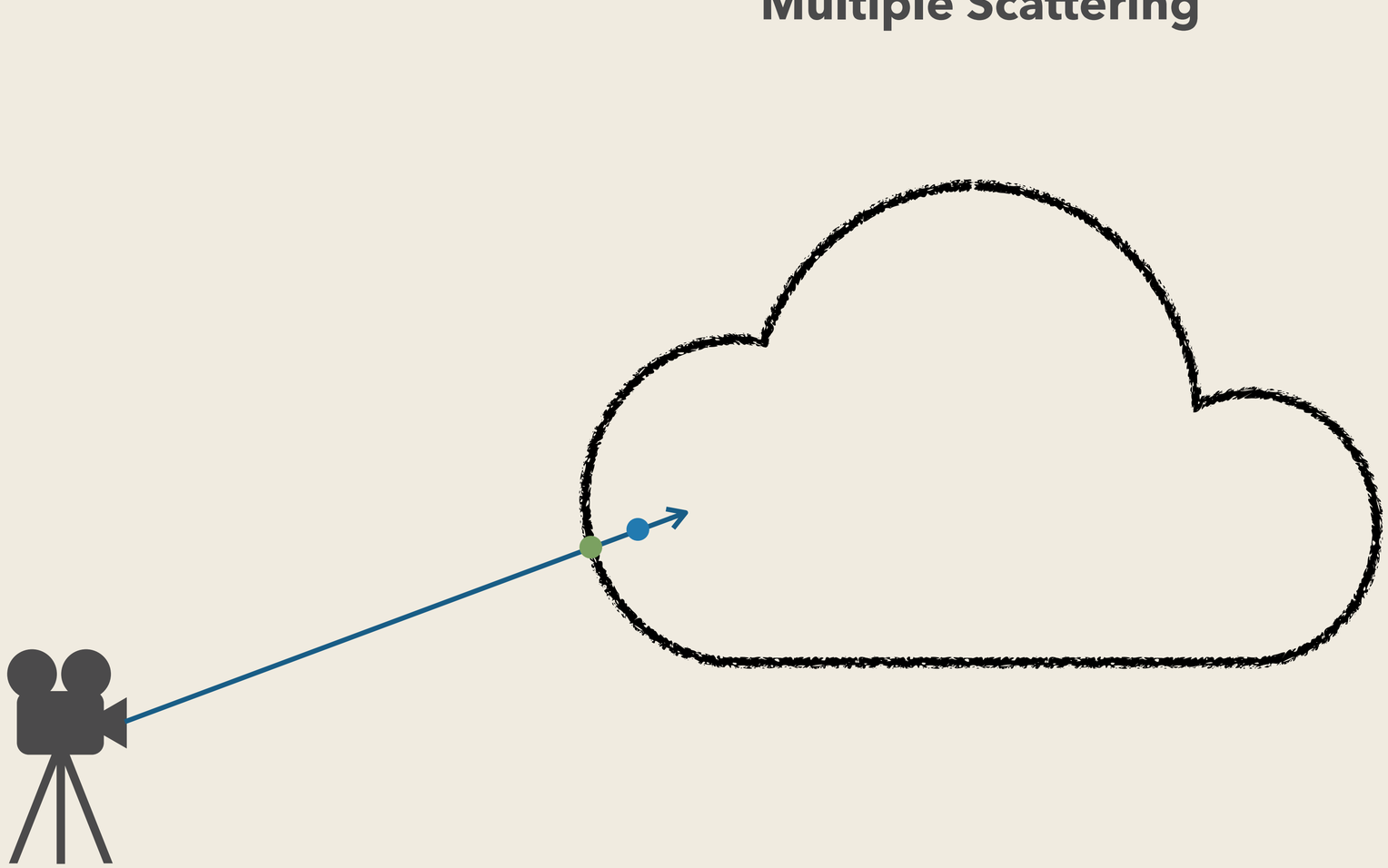
Heterogenous Participating Media

Multiple Scattering



Heterogenous Participating Media

Multiple Scattering



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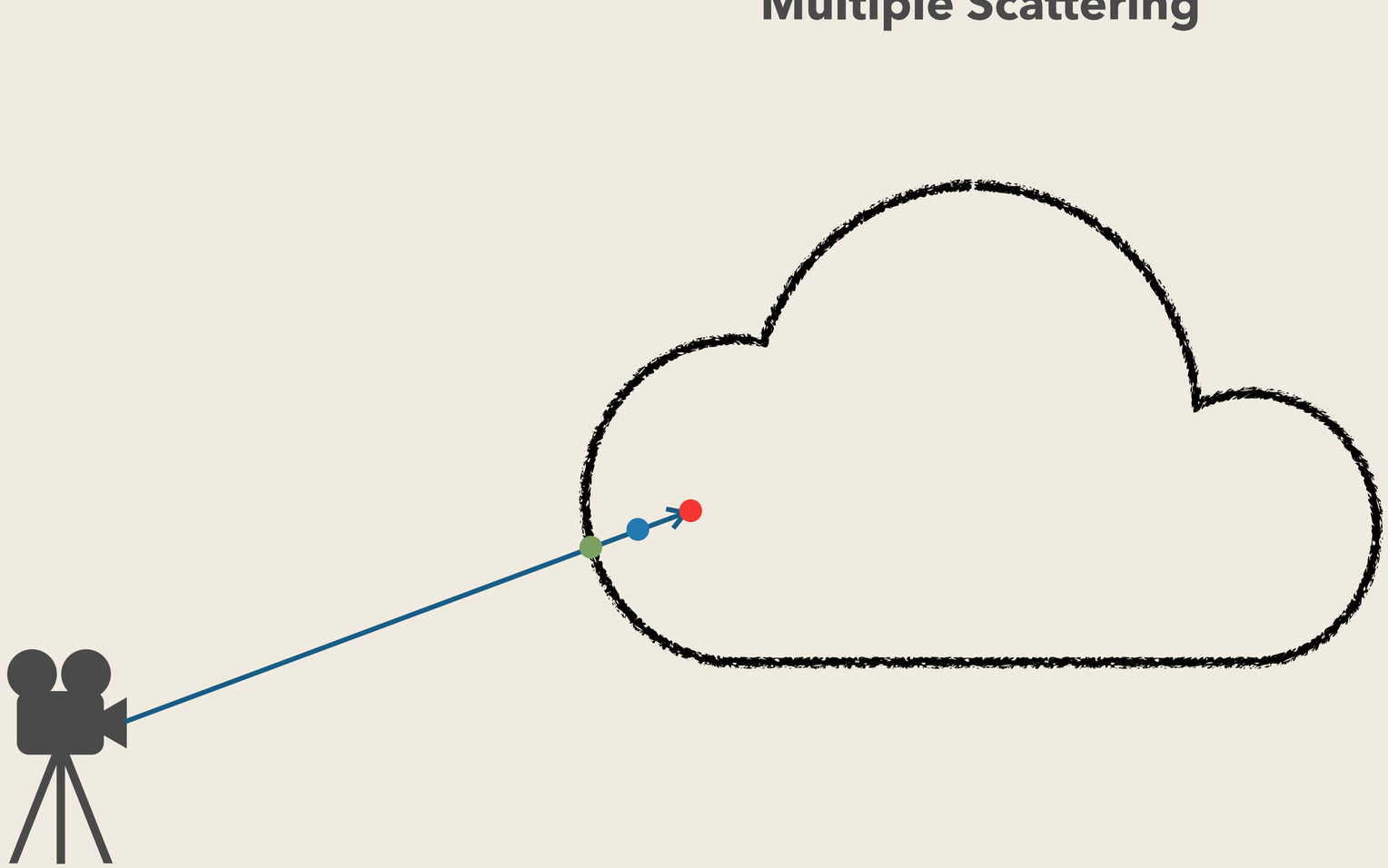

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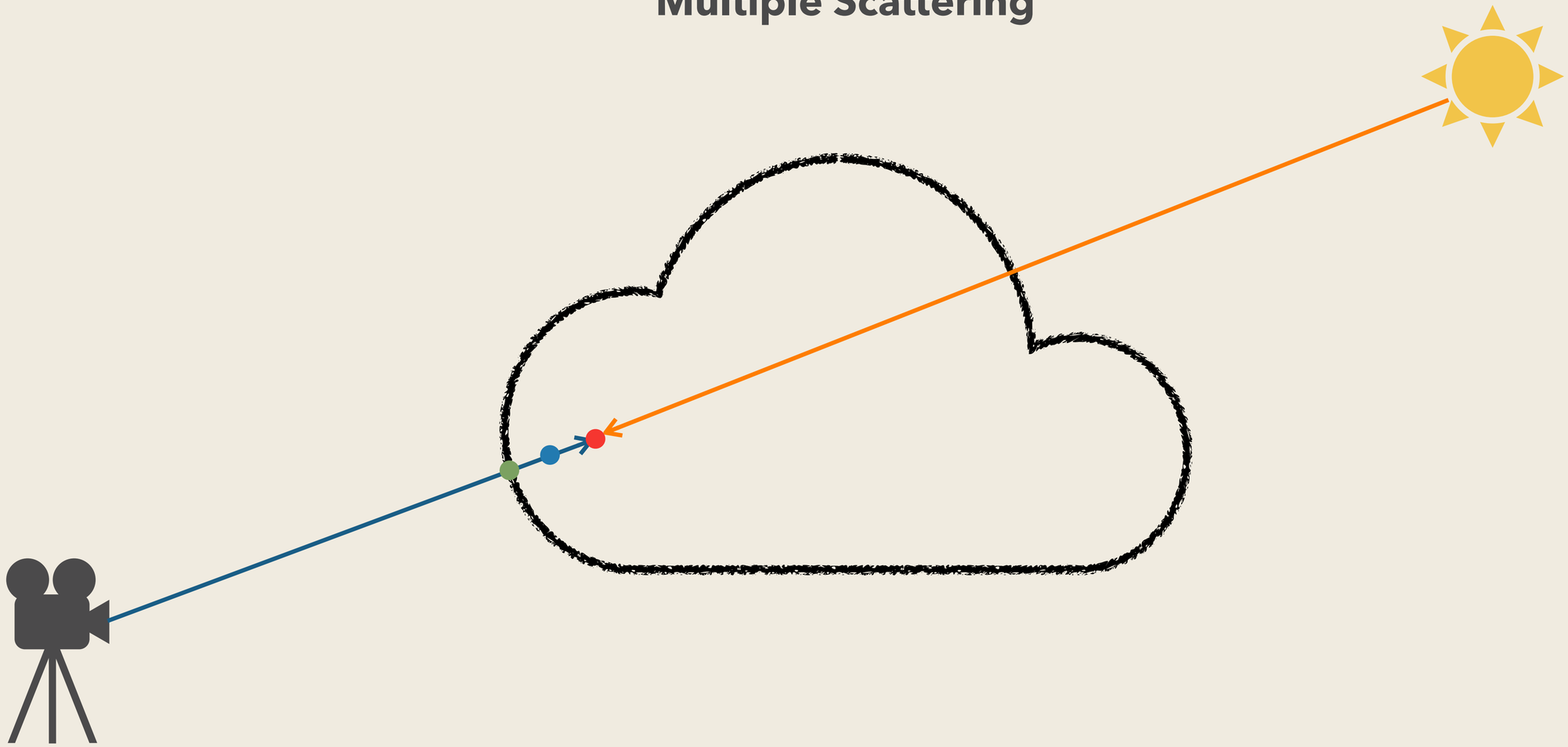
Multiple Scattering



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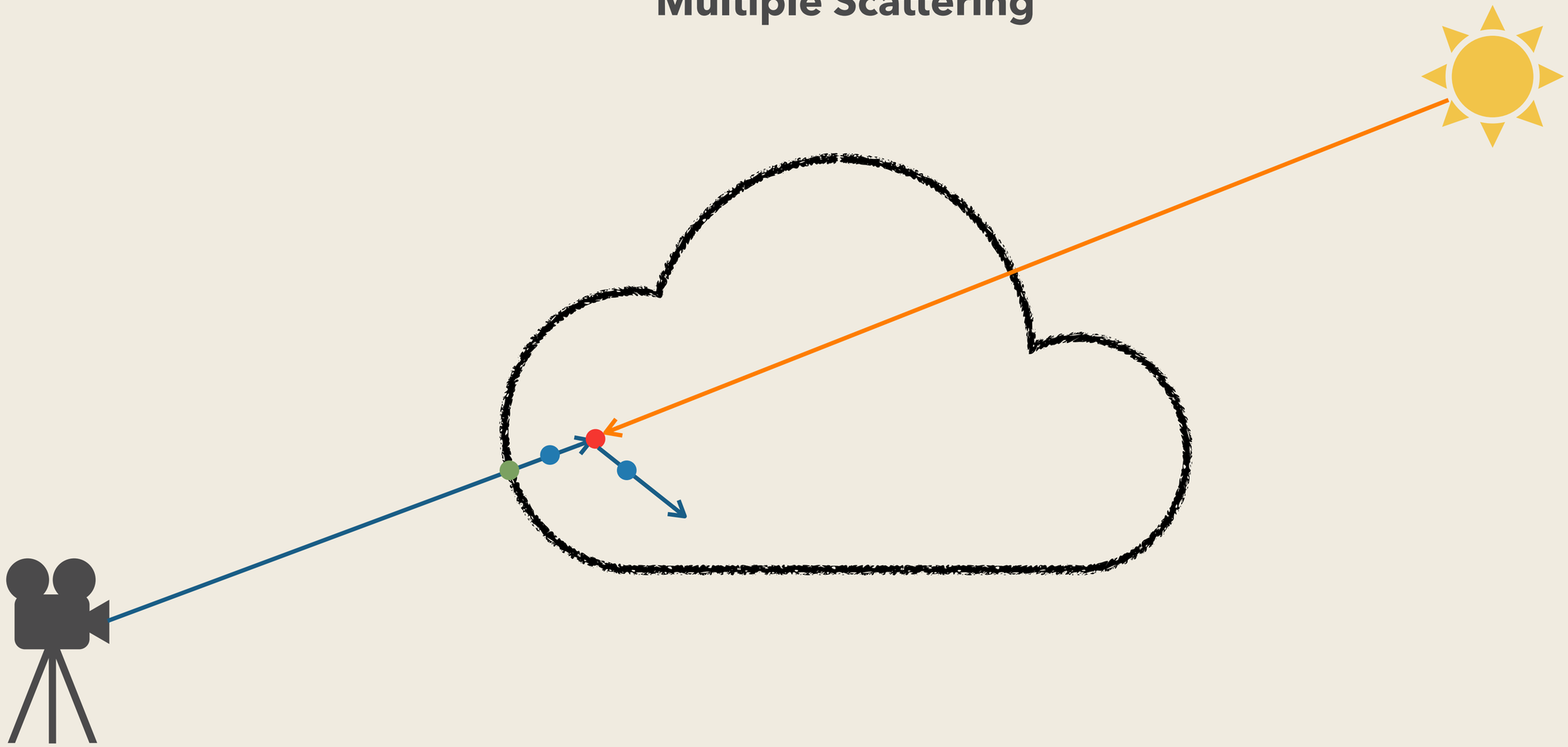

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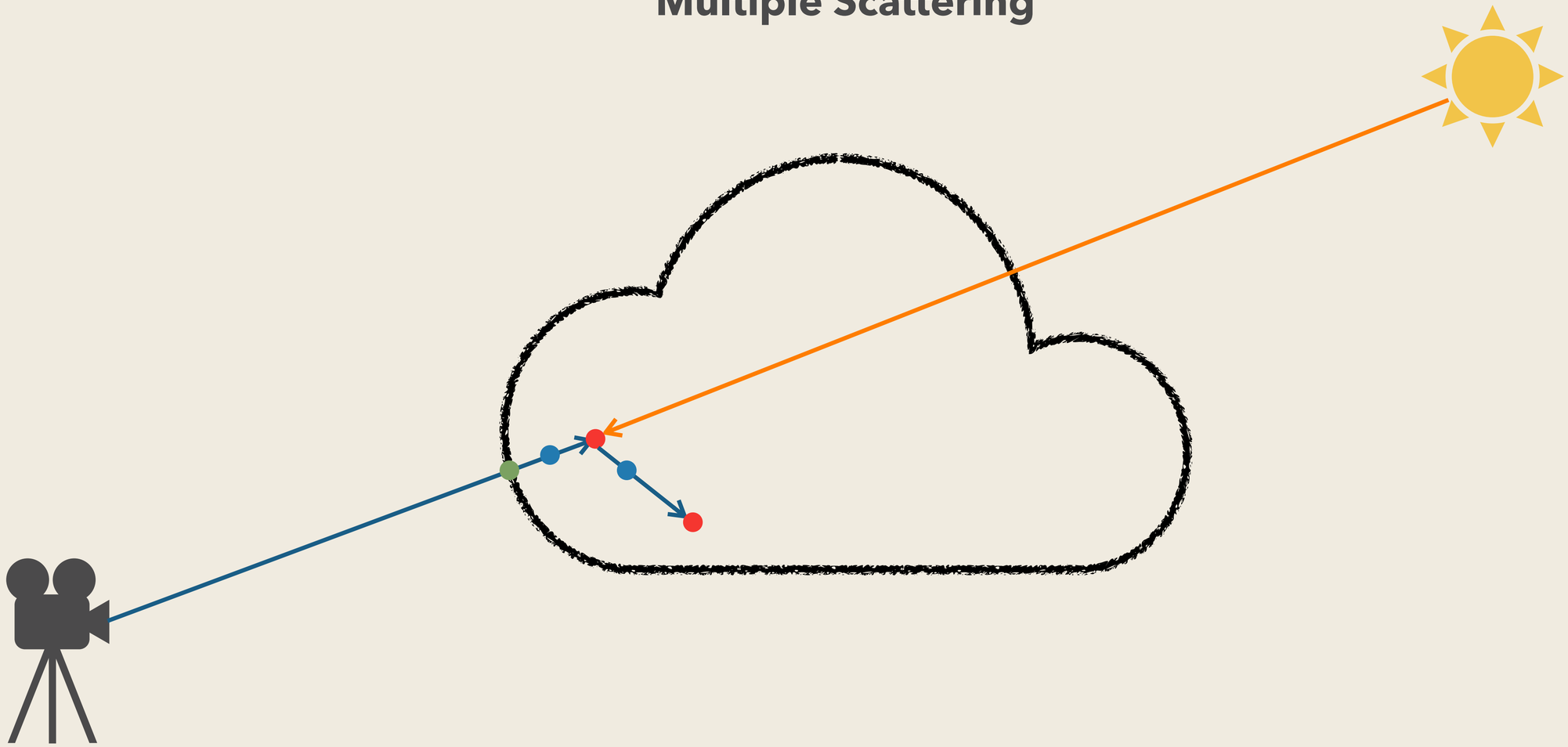

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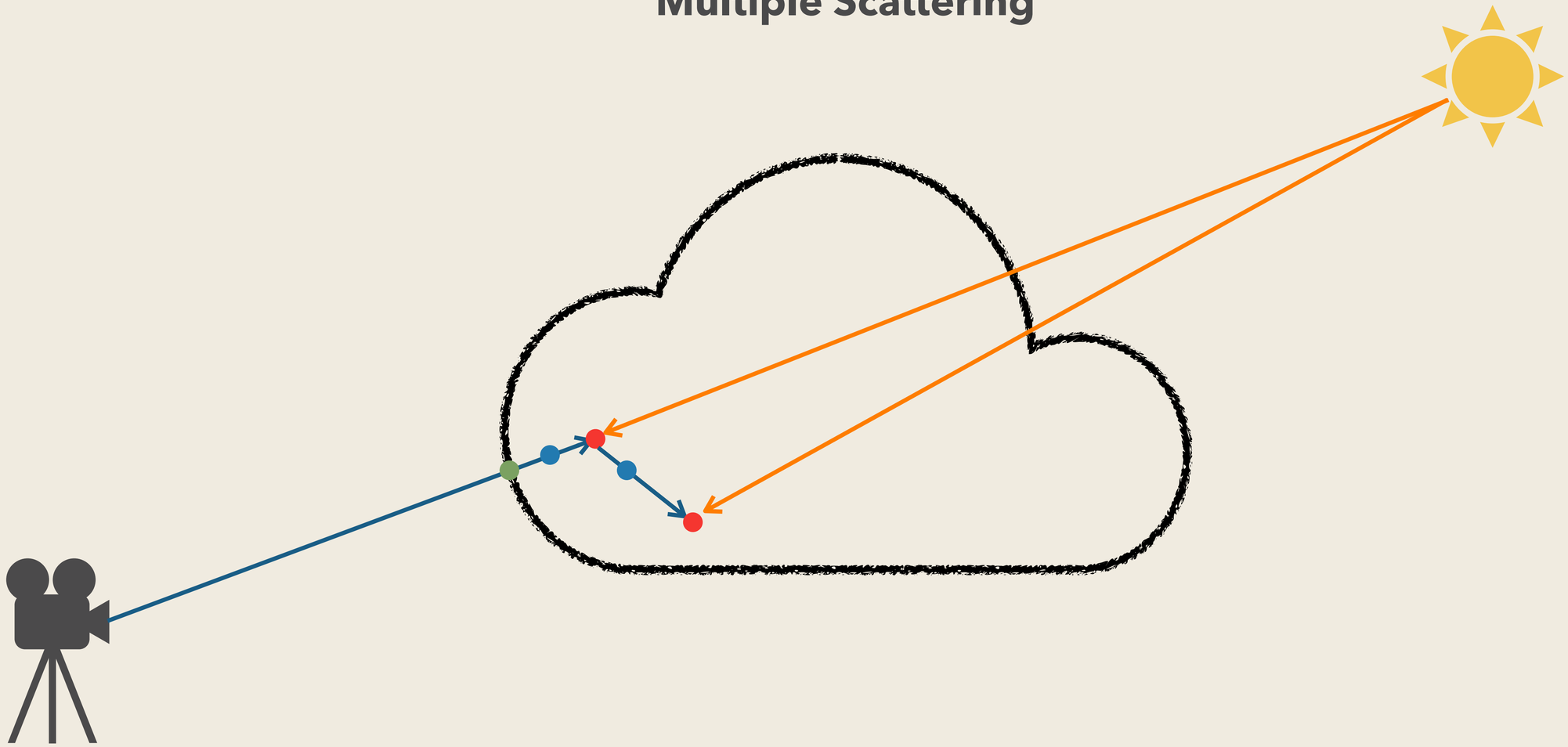
Multiple Scattering



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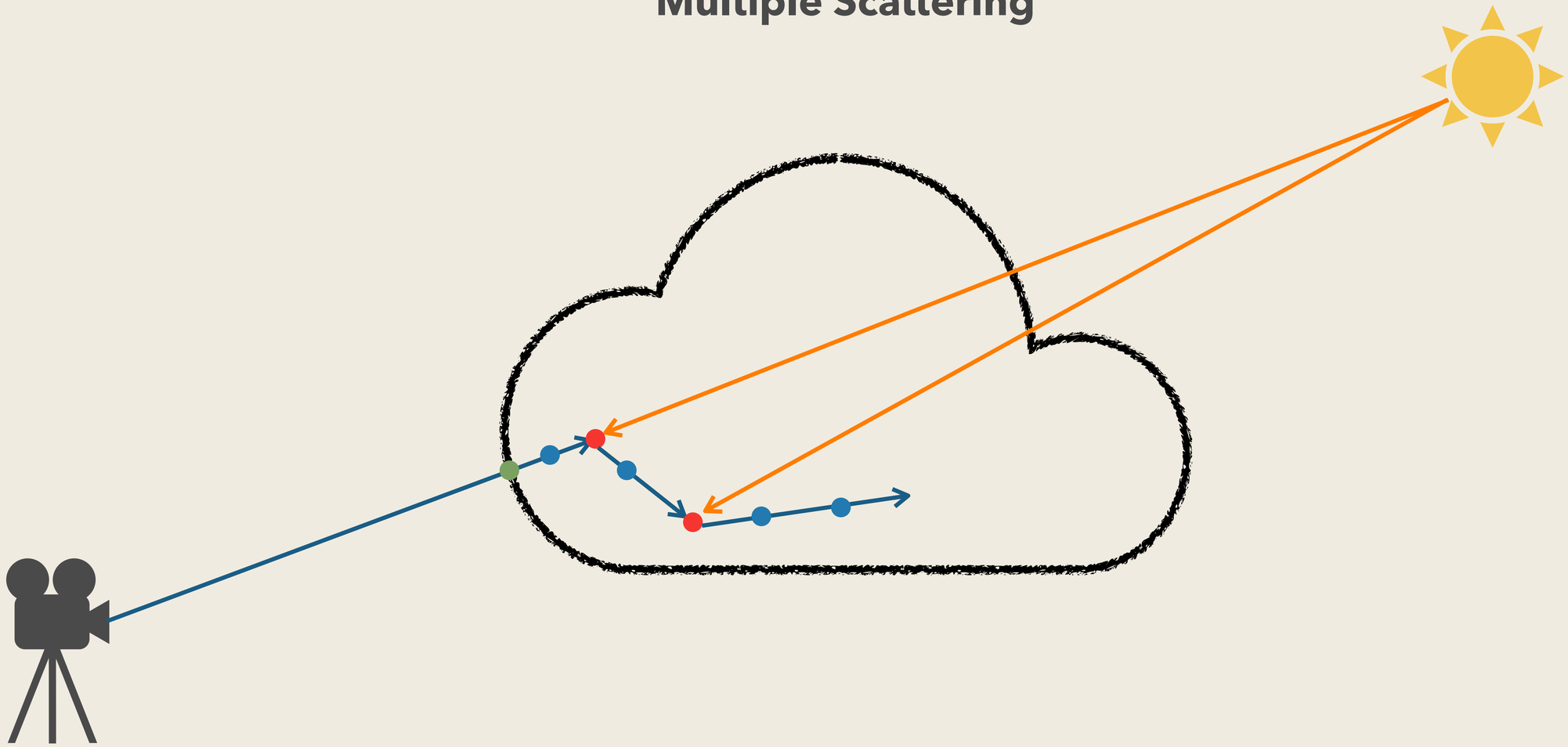
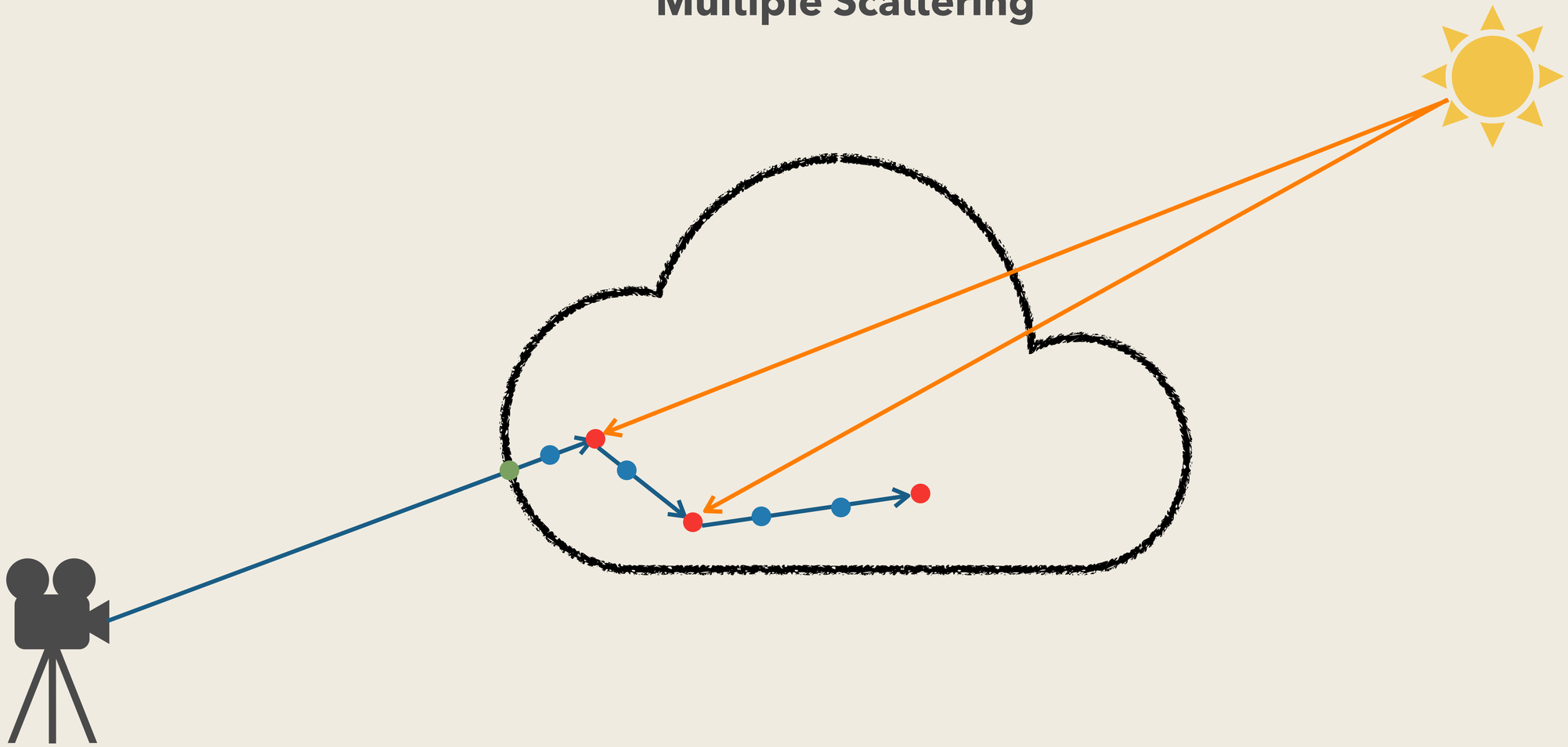
Multiple Scattering



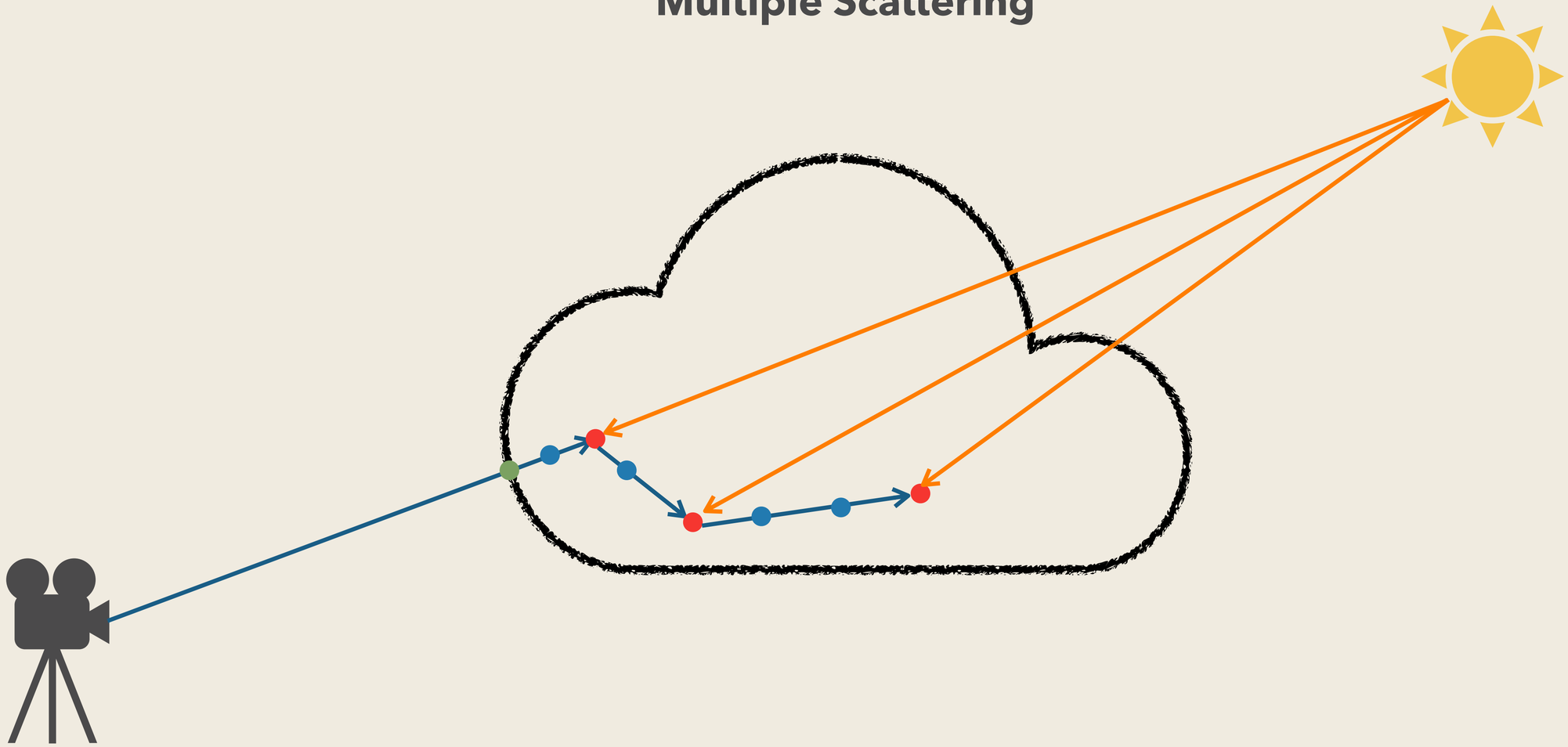
Heterogenous Participating Media

Multiple Scattering



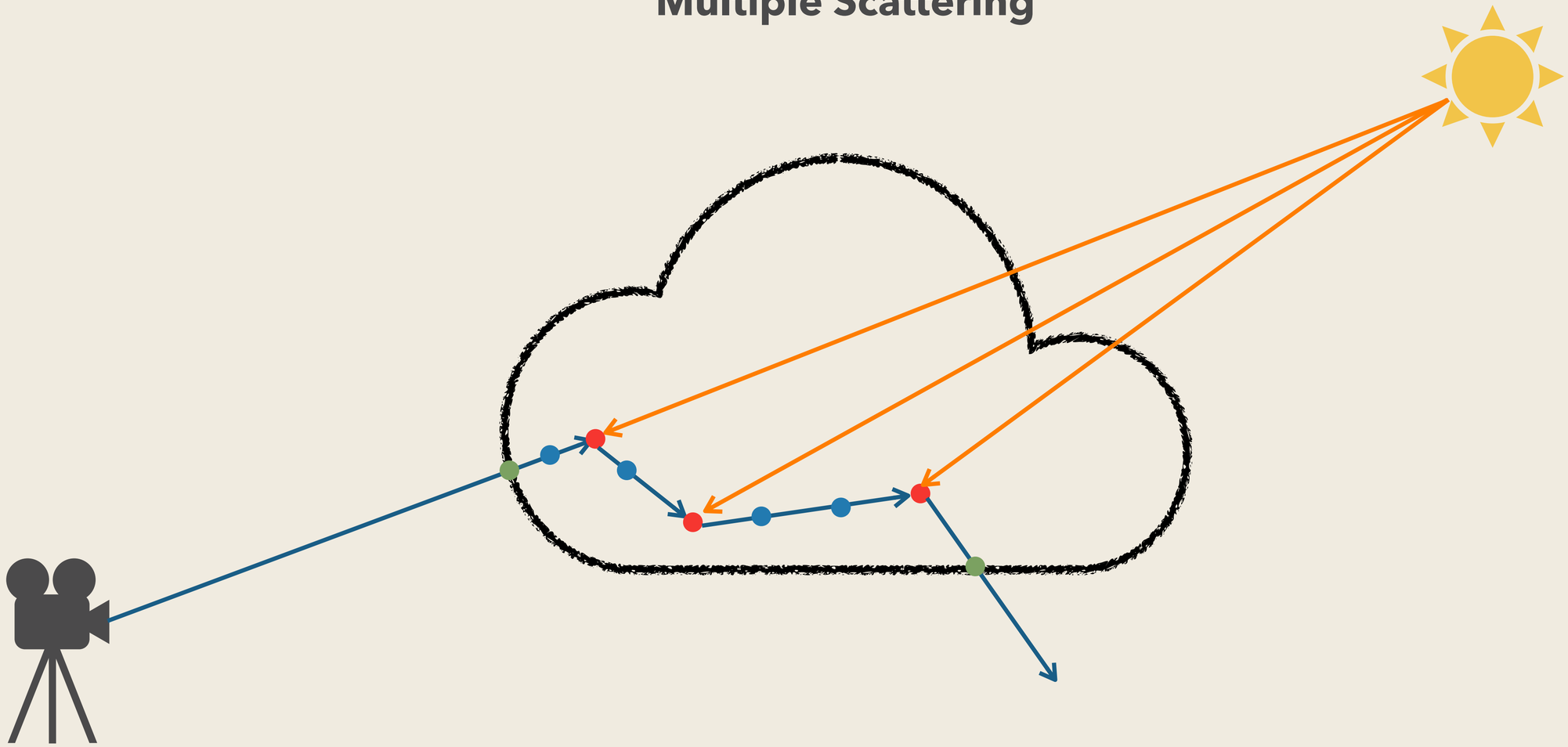
Heterogenous Participating Media

Multiple Scattering



Heterogenous Participating Media

Multiple Scattering



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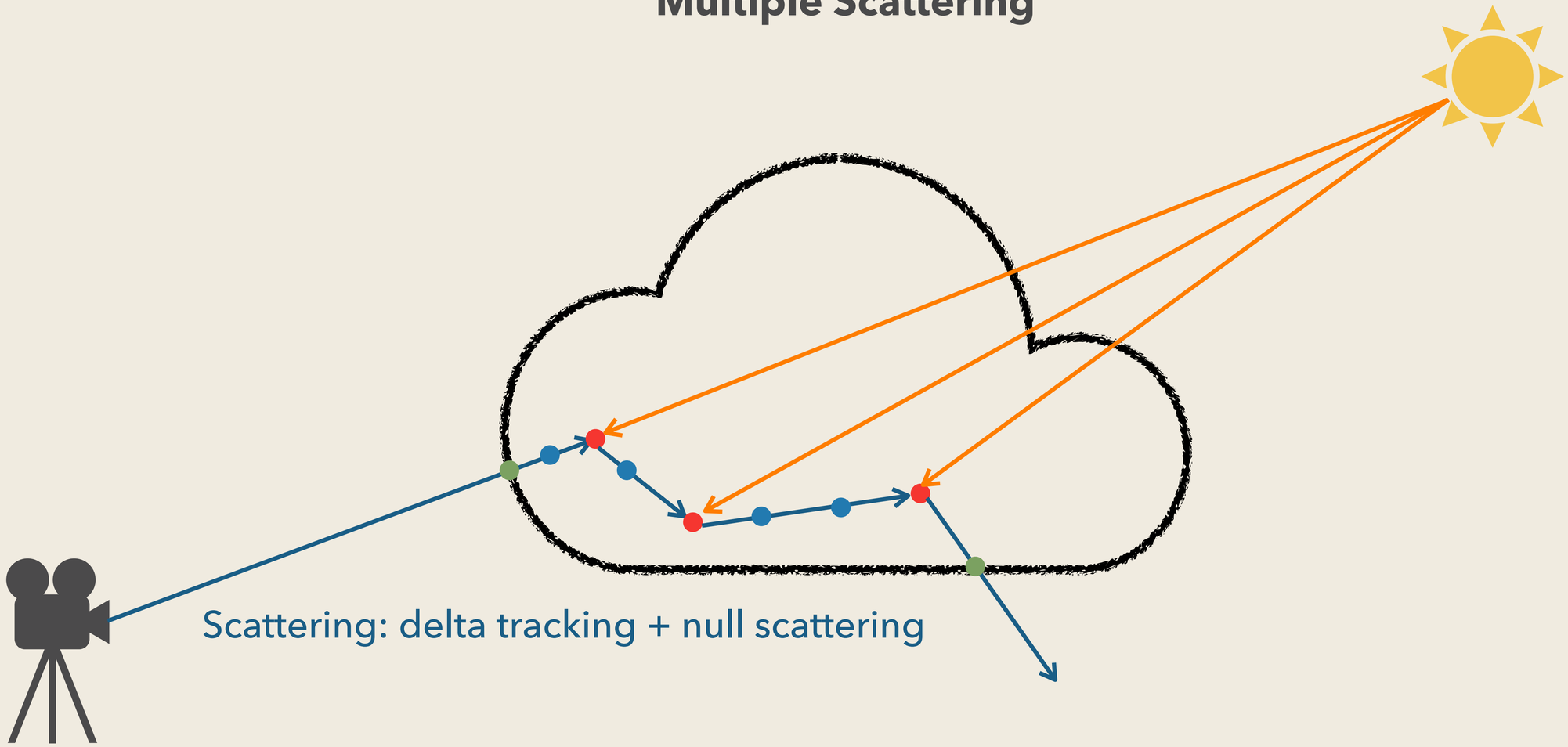

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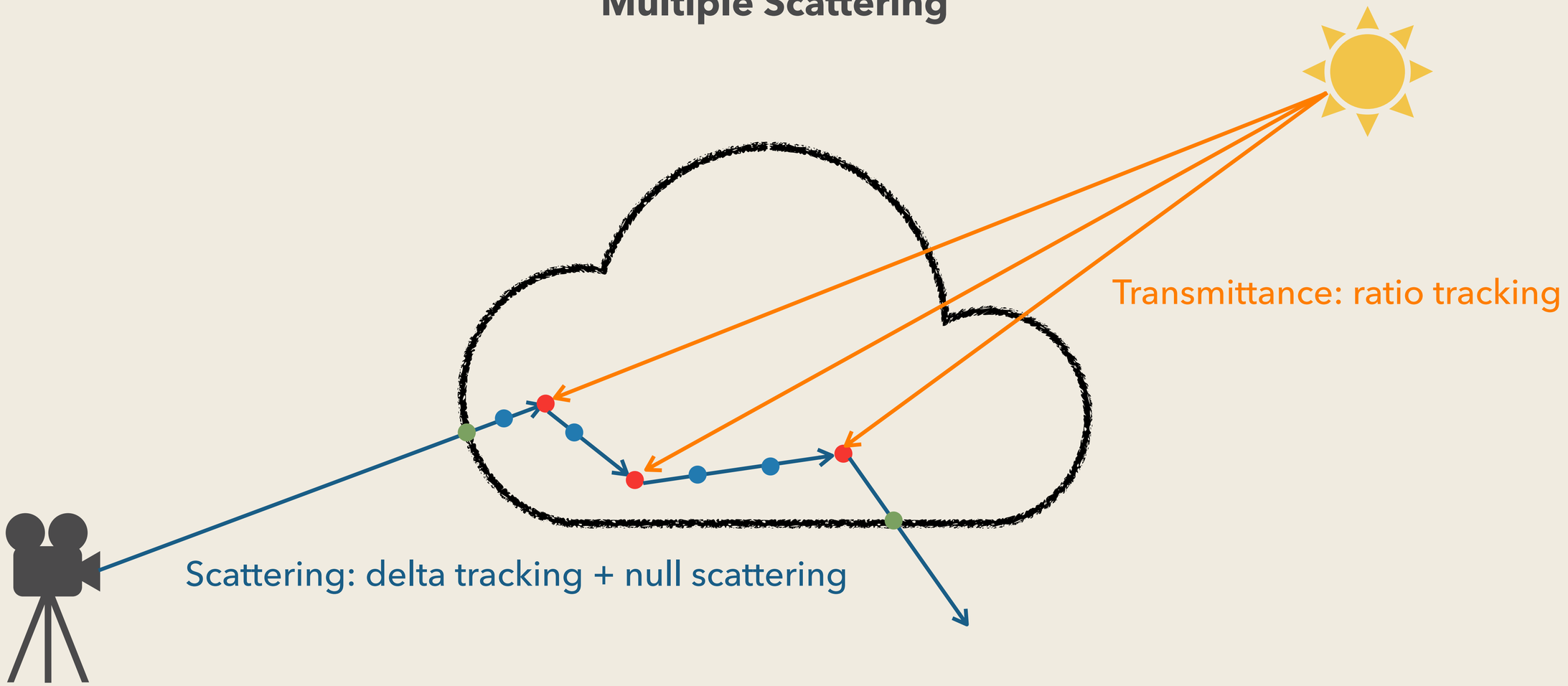
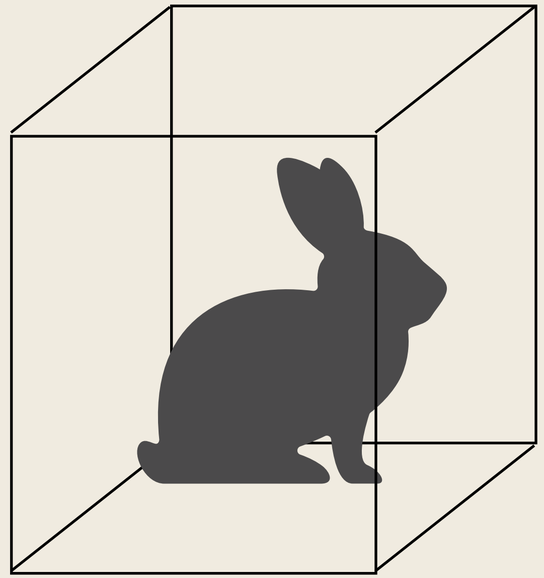


Heterogenous Participating Media

NanoVDB Grid

Heterogenous Participating Media

NanoVDB Grid

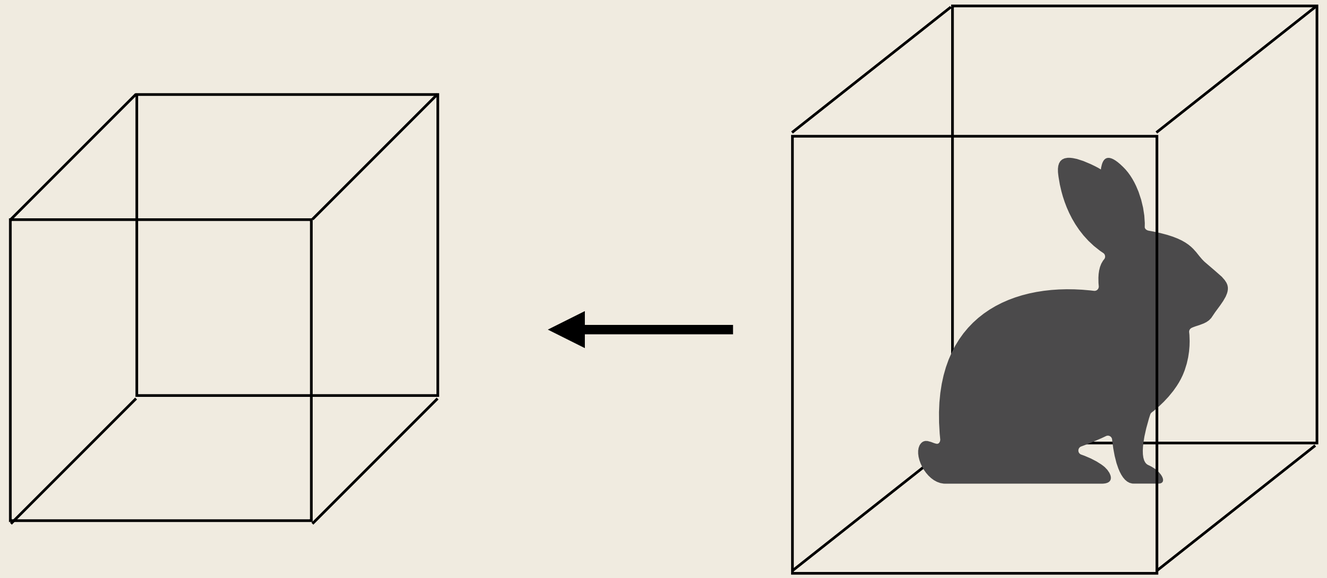


NanoVDB

```
grid→worldBBox()
```

Heterogenous Participating Media

NanoVDB Grid



Unit Cube

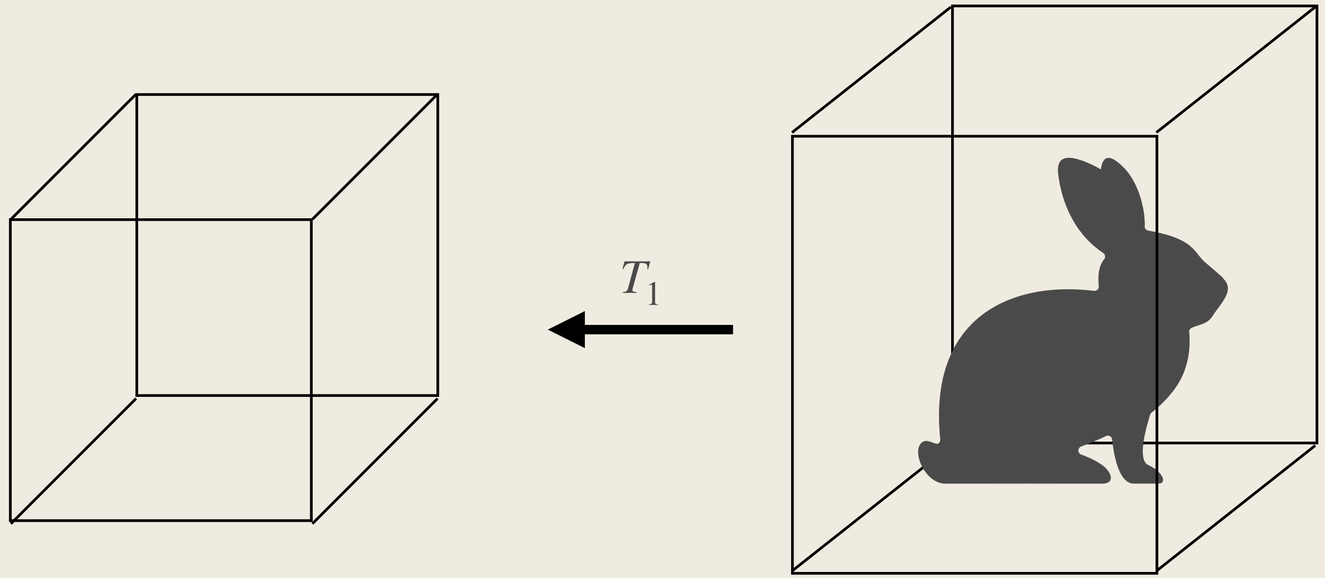
$[-1,1]^3$

NanoVDB

`grid→worldBBox()`

Heterogenous Participating Media

NanoVDB Grid



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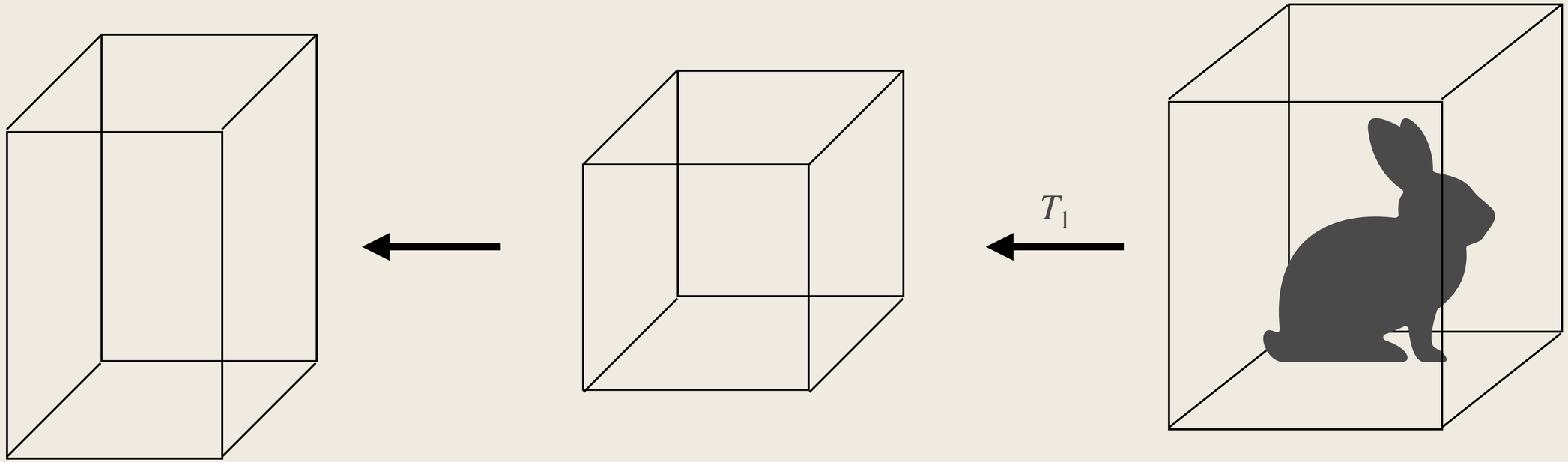

User-defined
Bounding Box

`Transform toWorld`

Unit Cube

$[-1,1]^3$

NanoVDB

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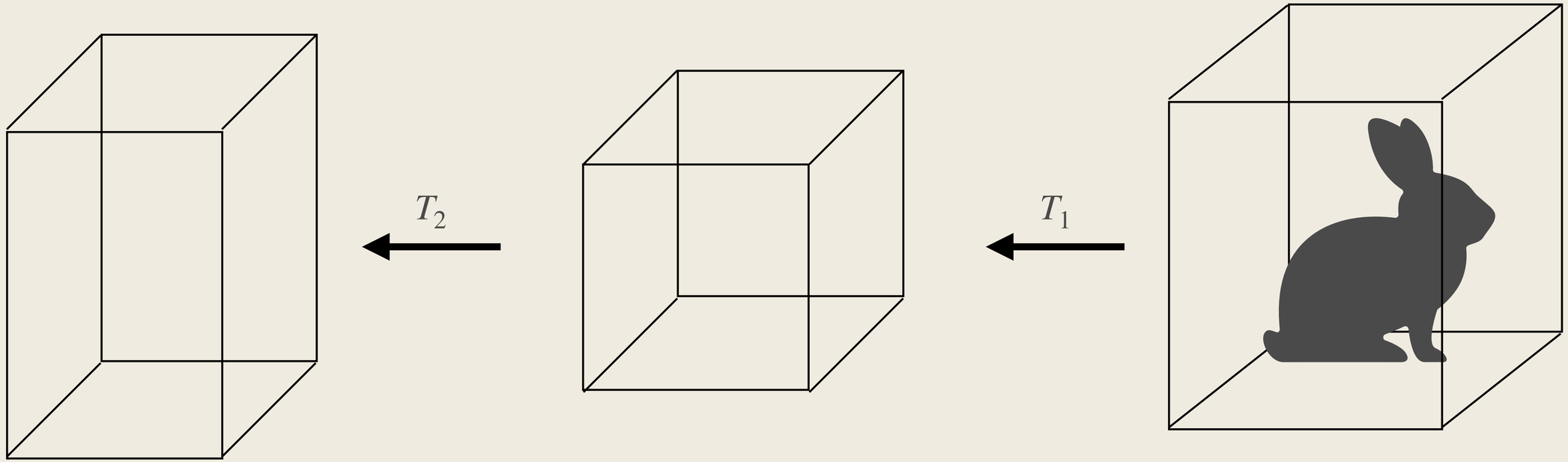
Heterogenous Participating Media

NanoVDB Grid



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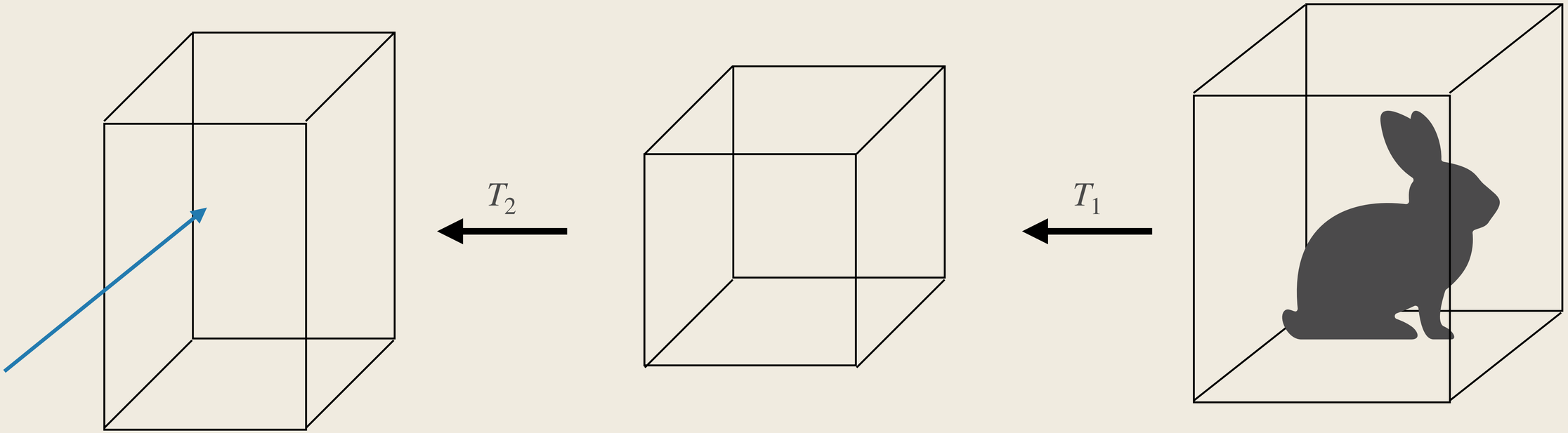

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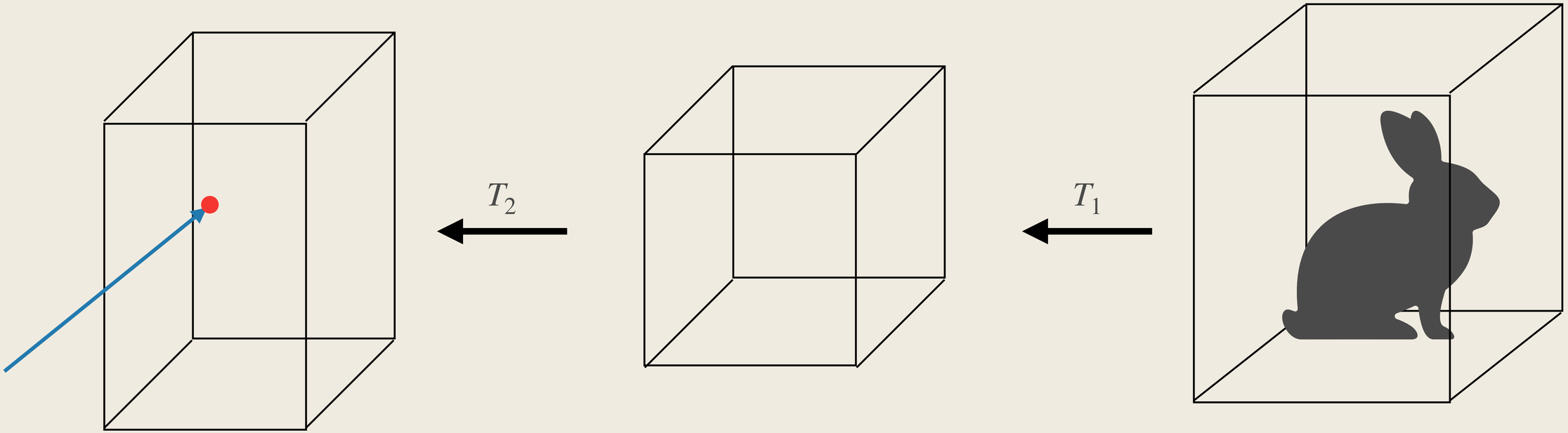

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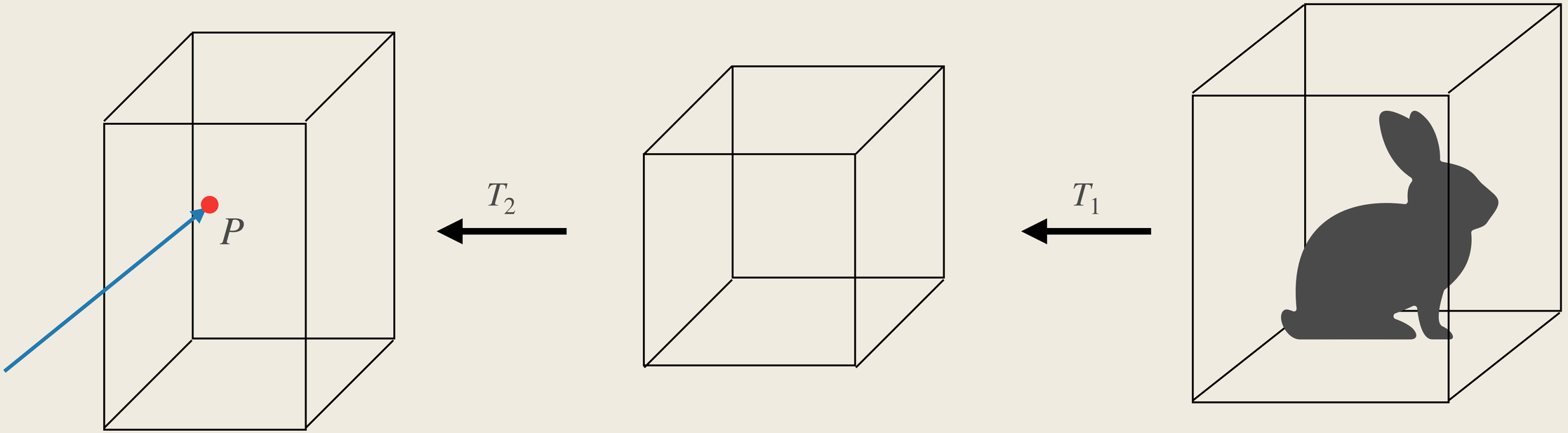

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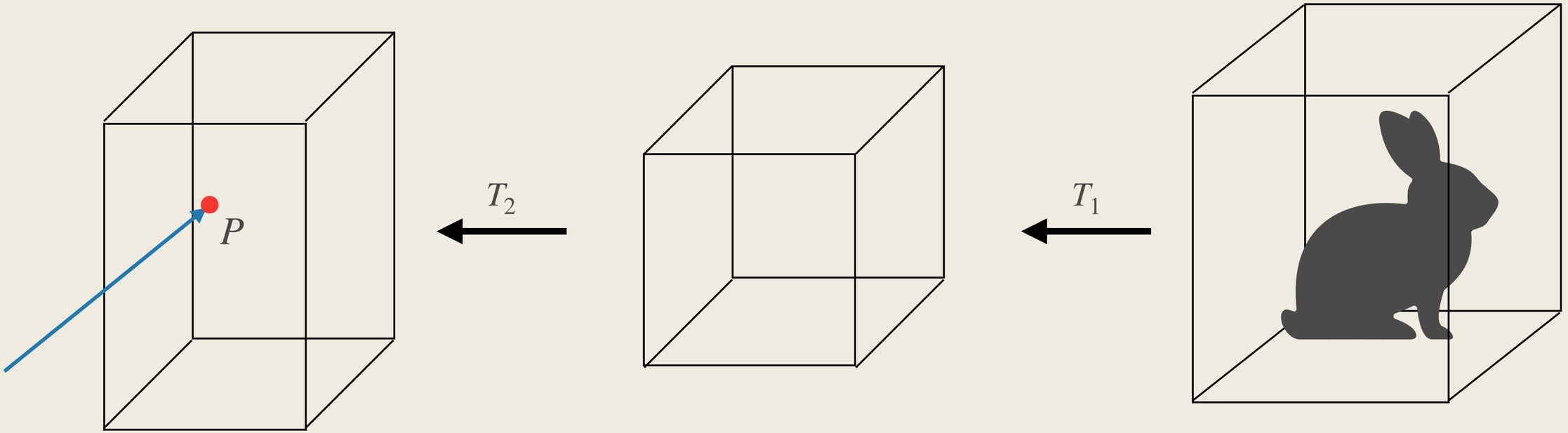
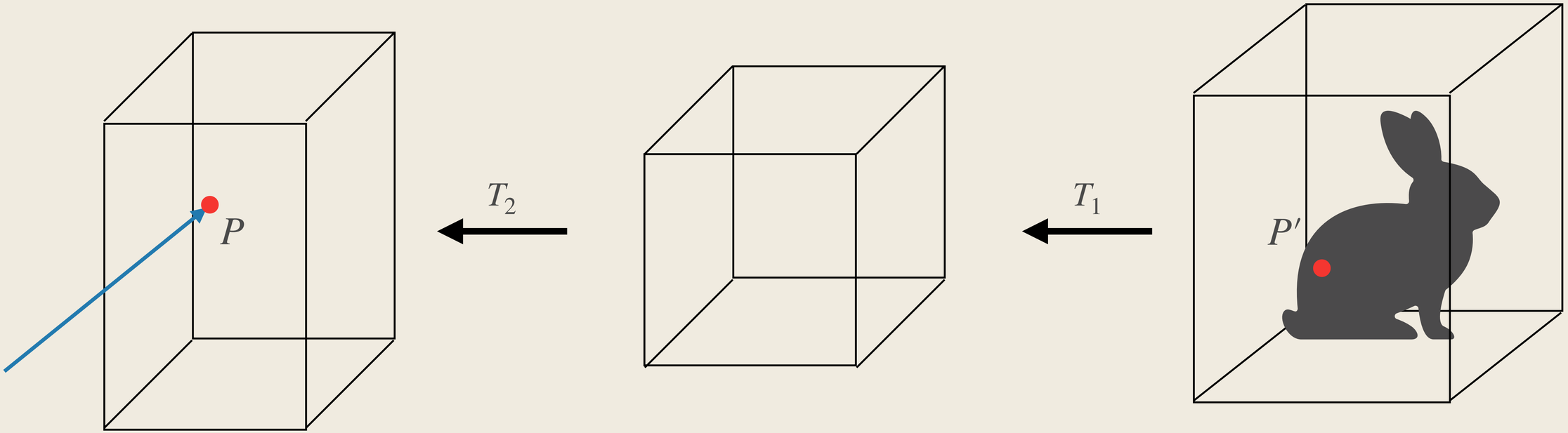


$$P' = T_1^{-1} \cdot T_2^{-1} \cdot P$$

Heterogenous Participating Media

NanoVDB Grid



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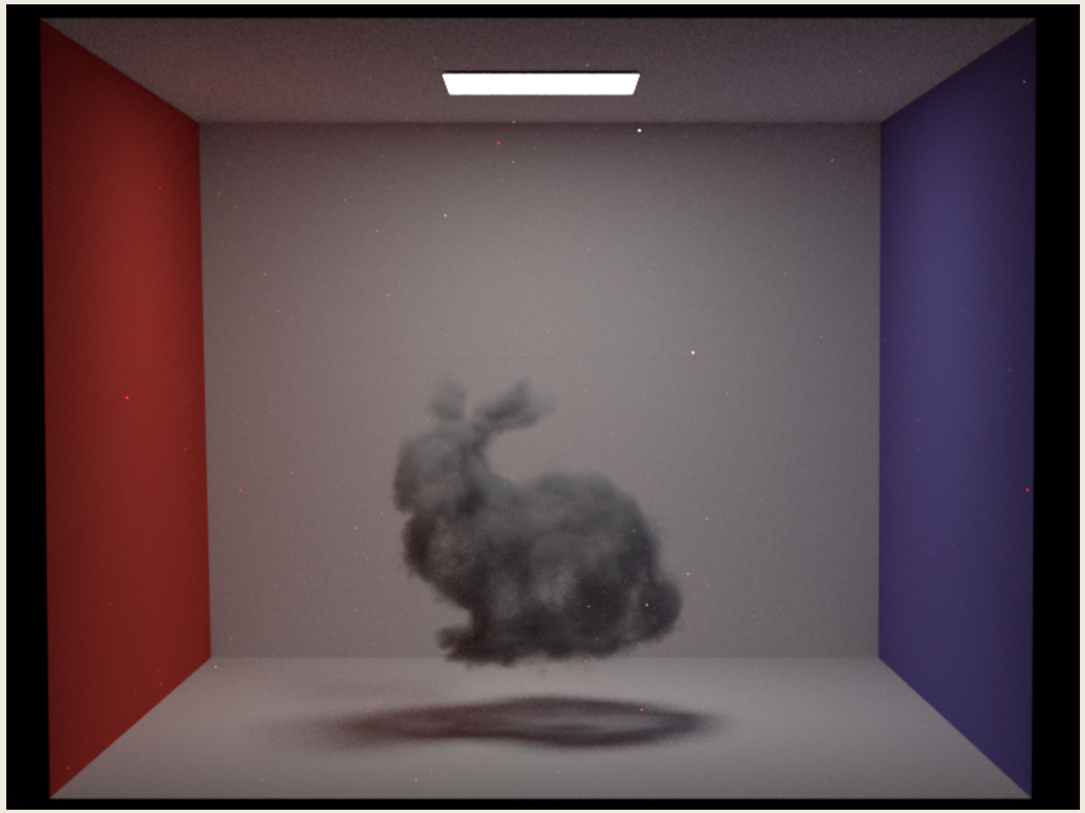
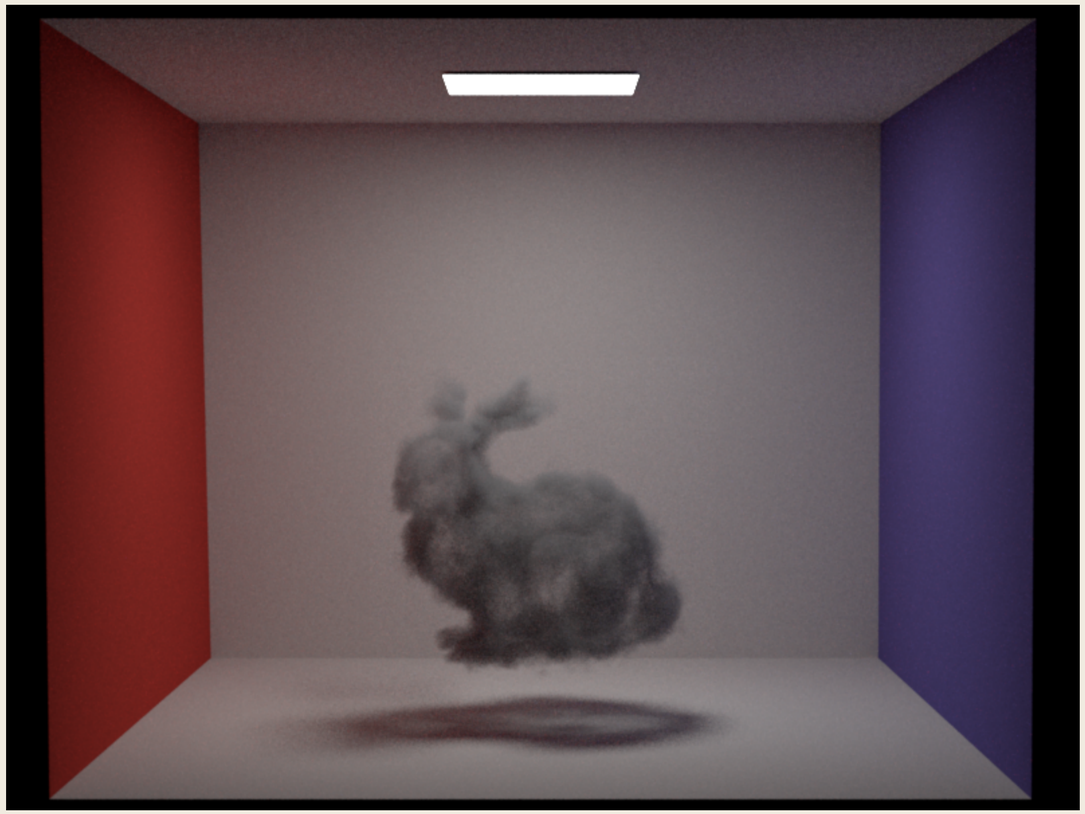
Heterogenous Participating Media

Results

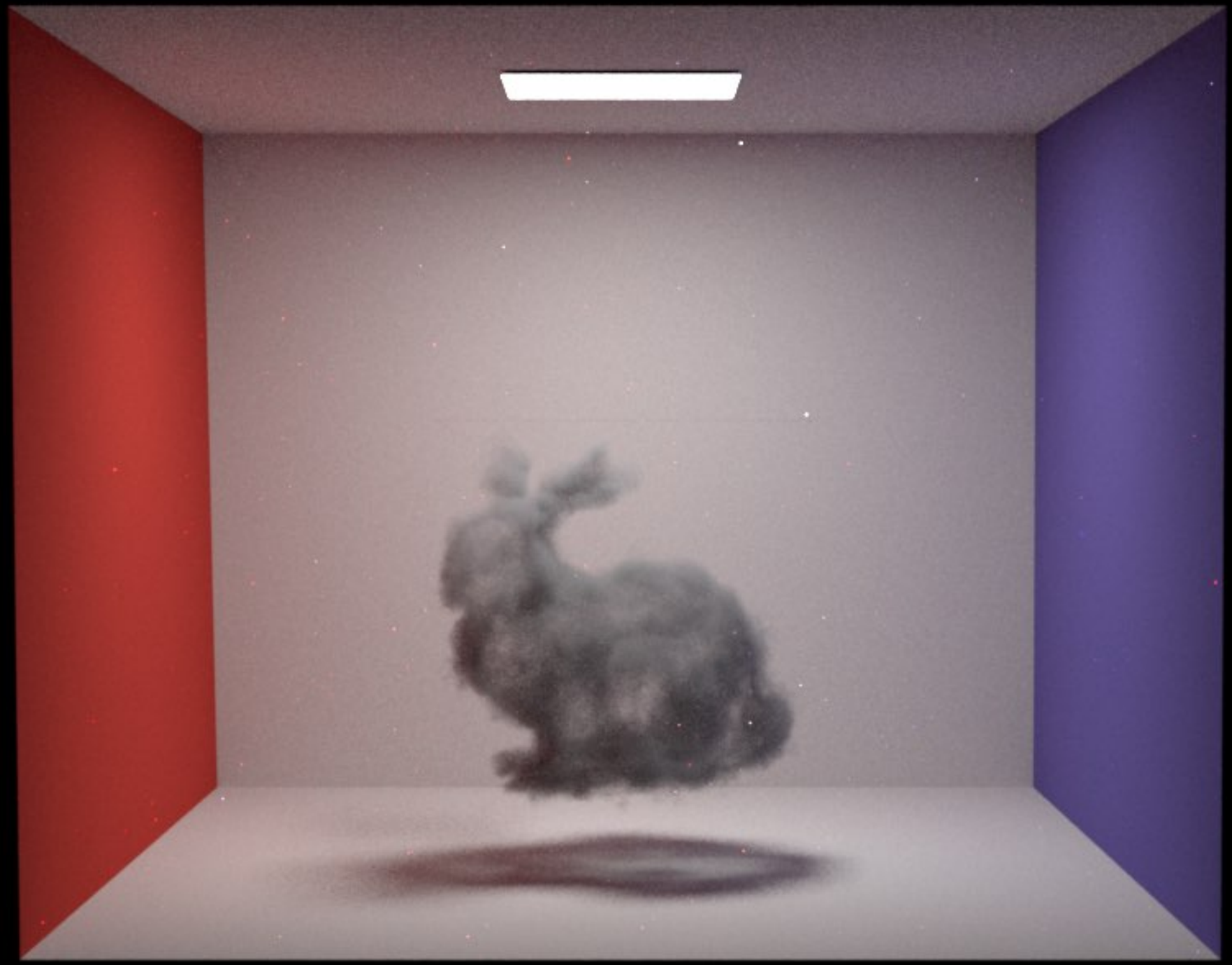


Mitsuba 3

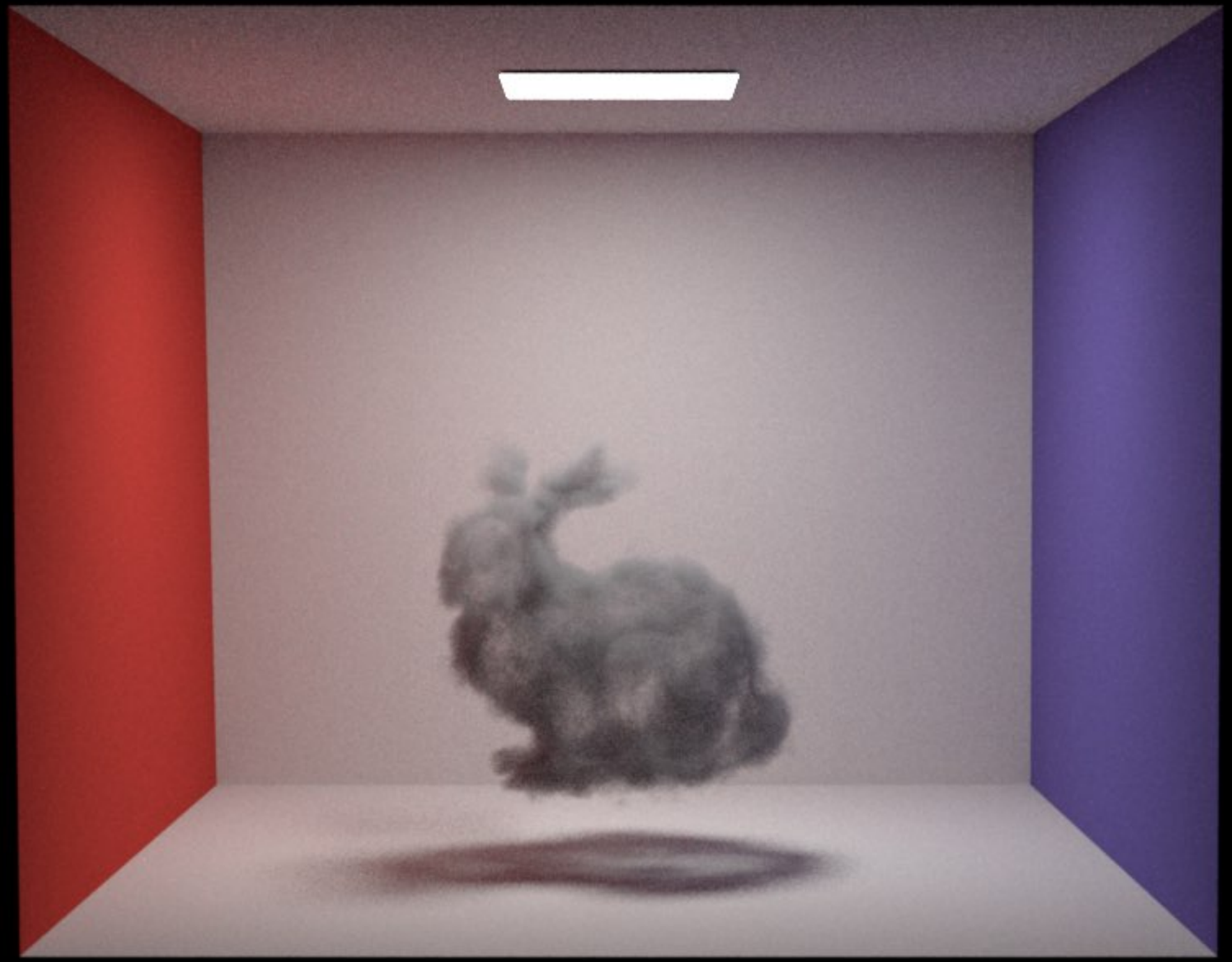


Ours

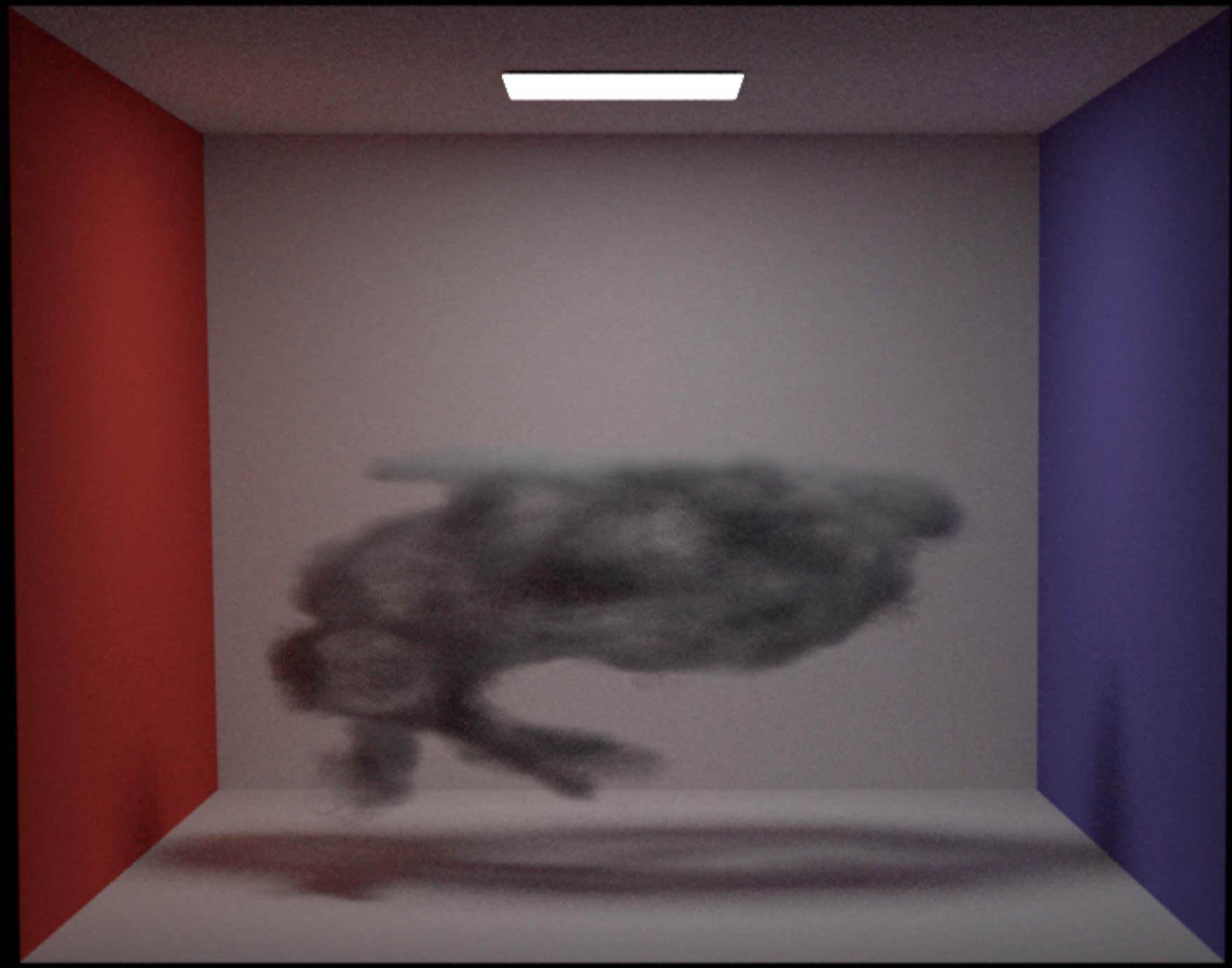
Mitsuba 3



Ours

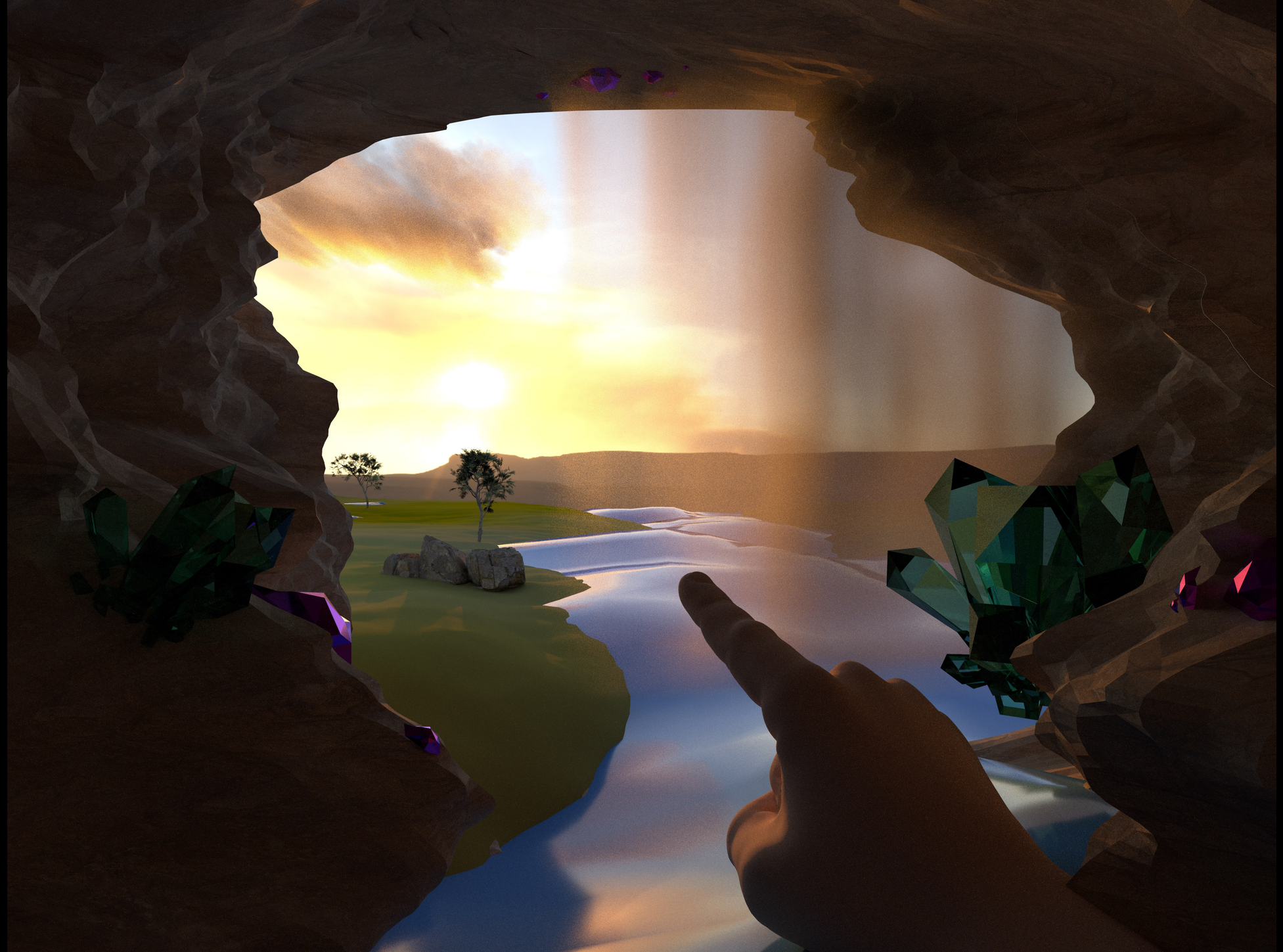


Ours
rotate + scale

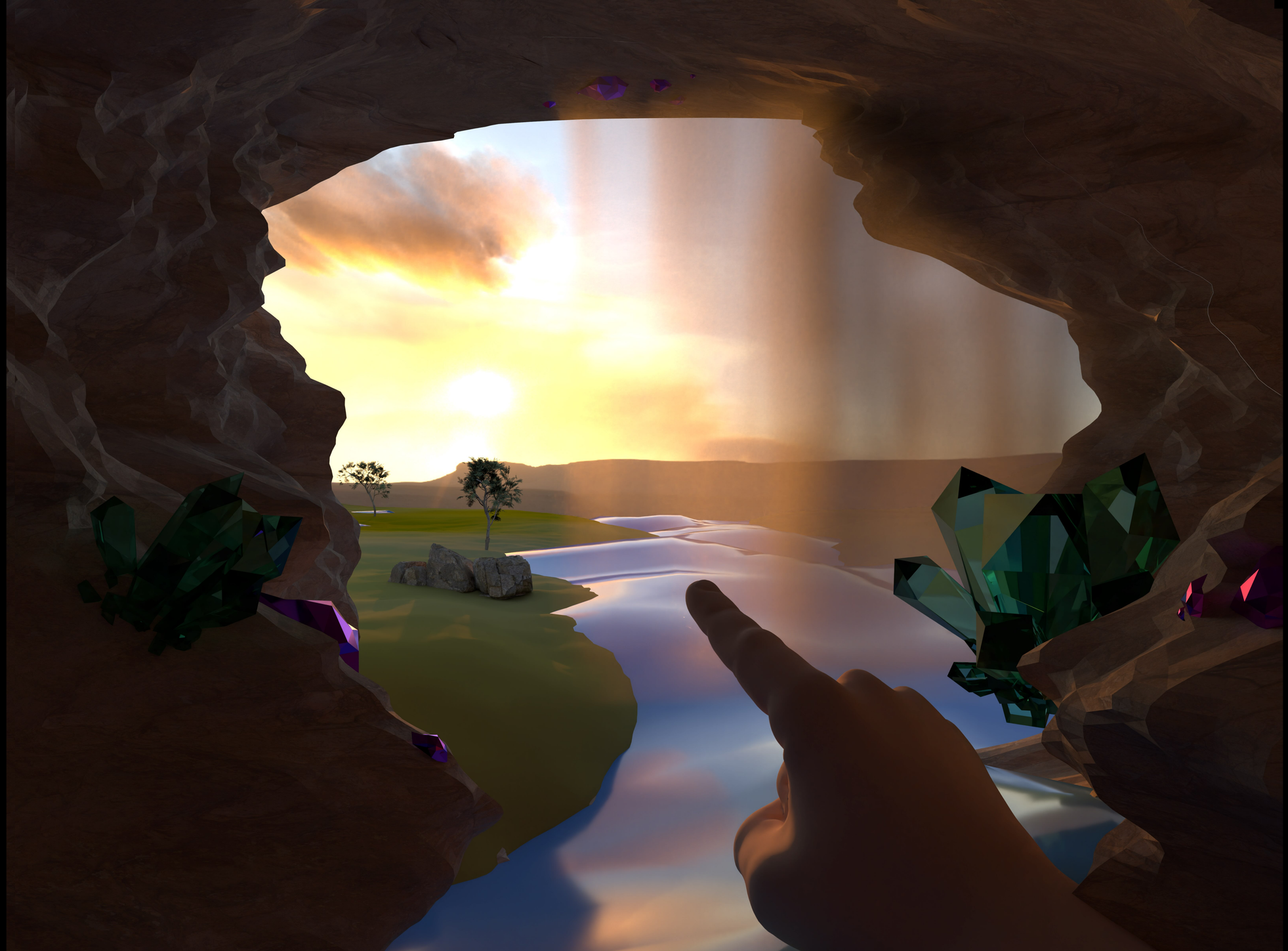


Final Image

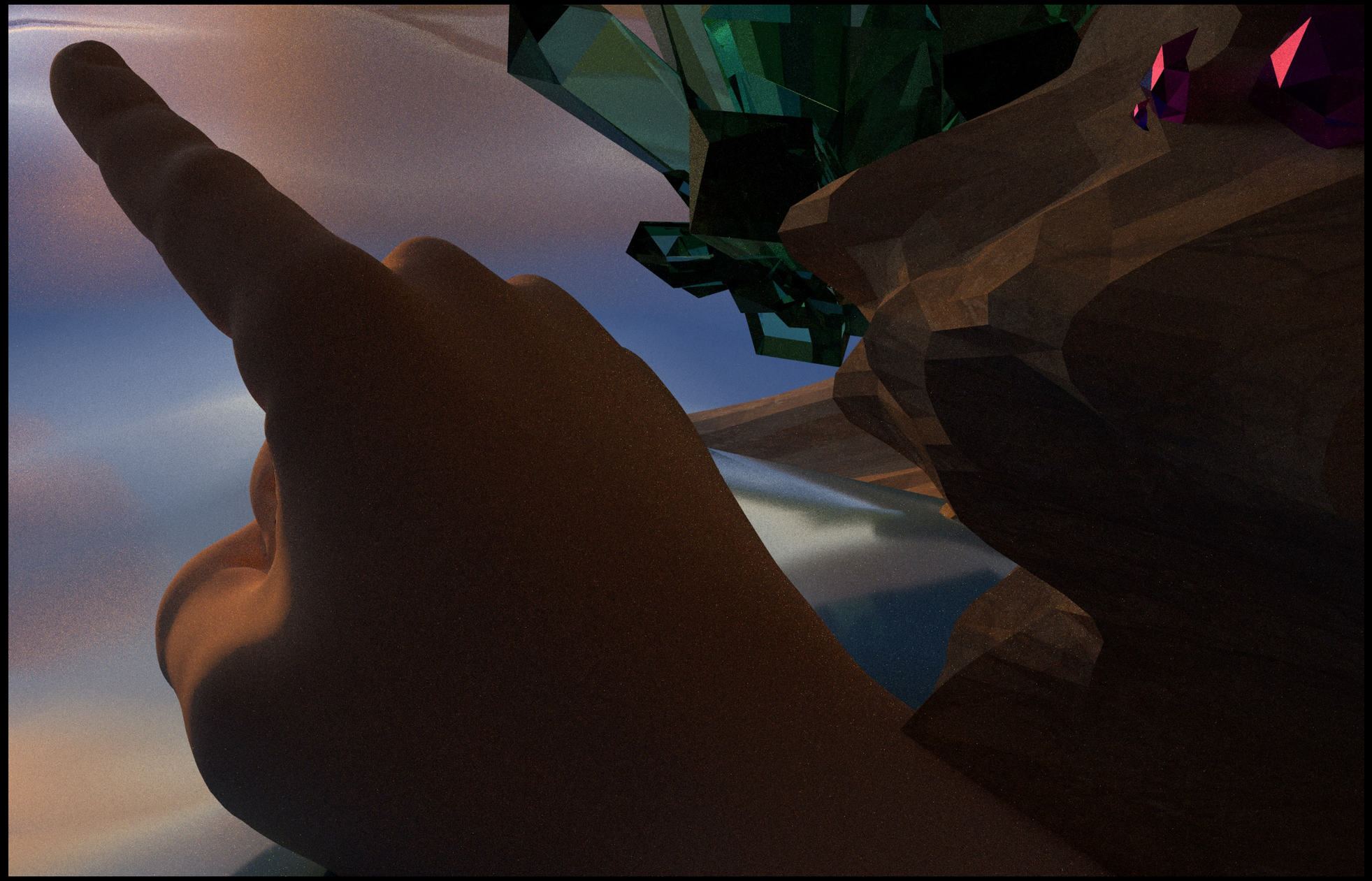
Original
32 spp



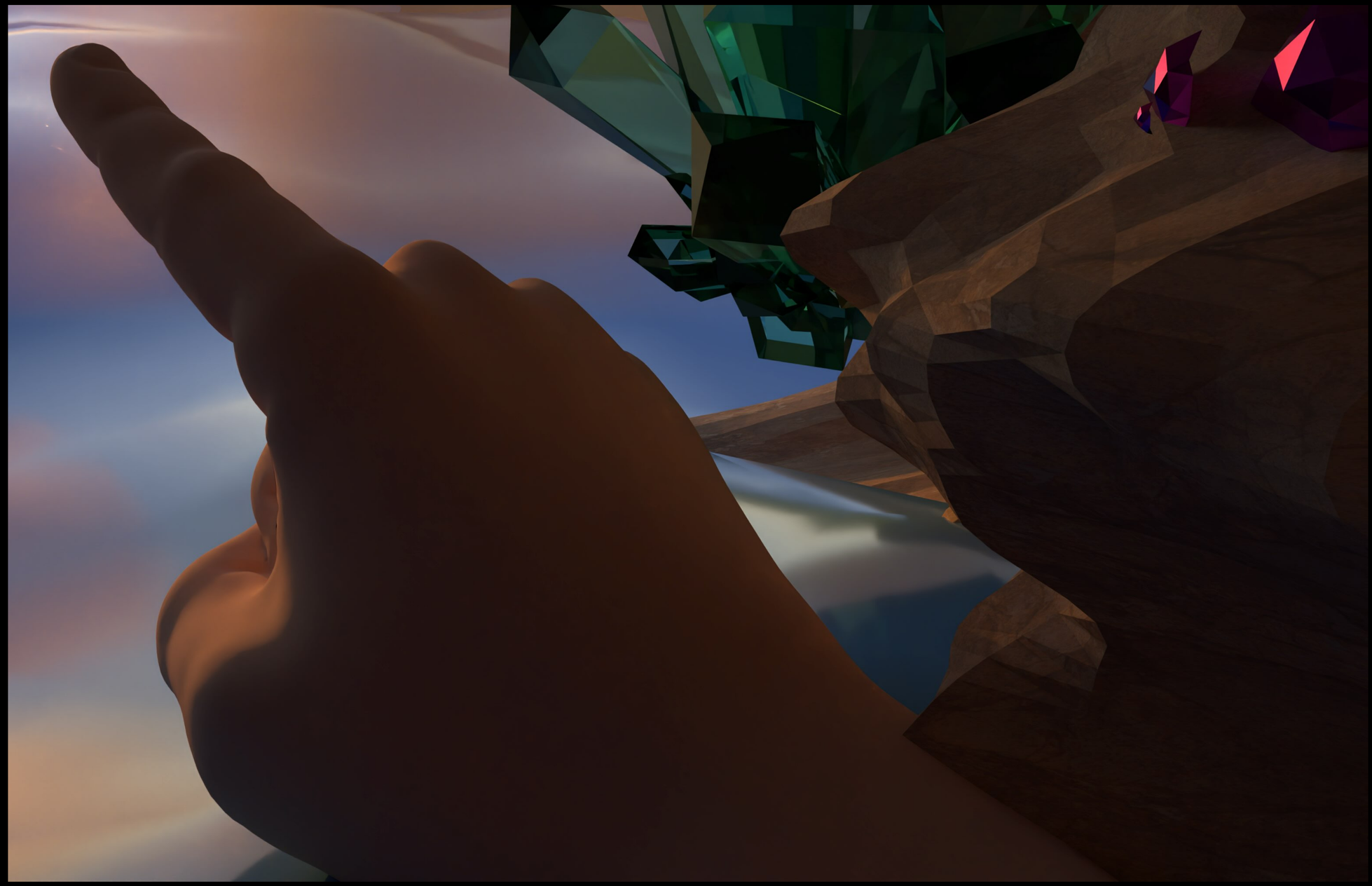
Denoised



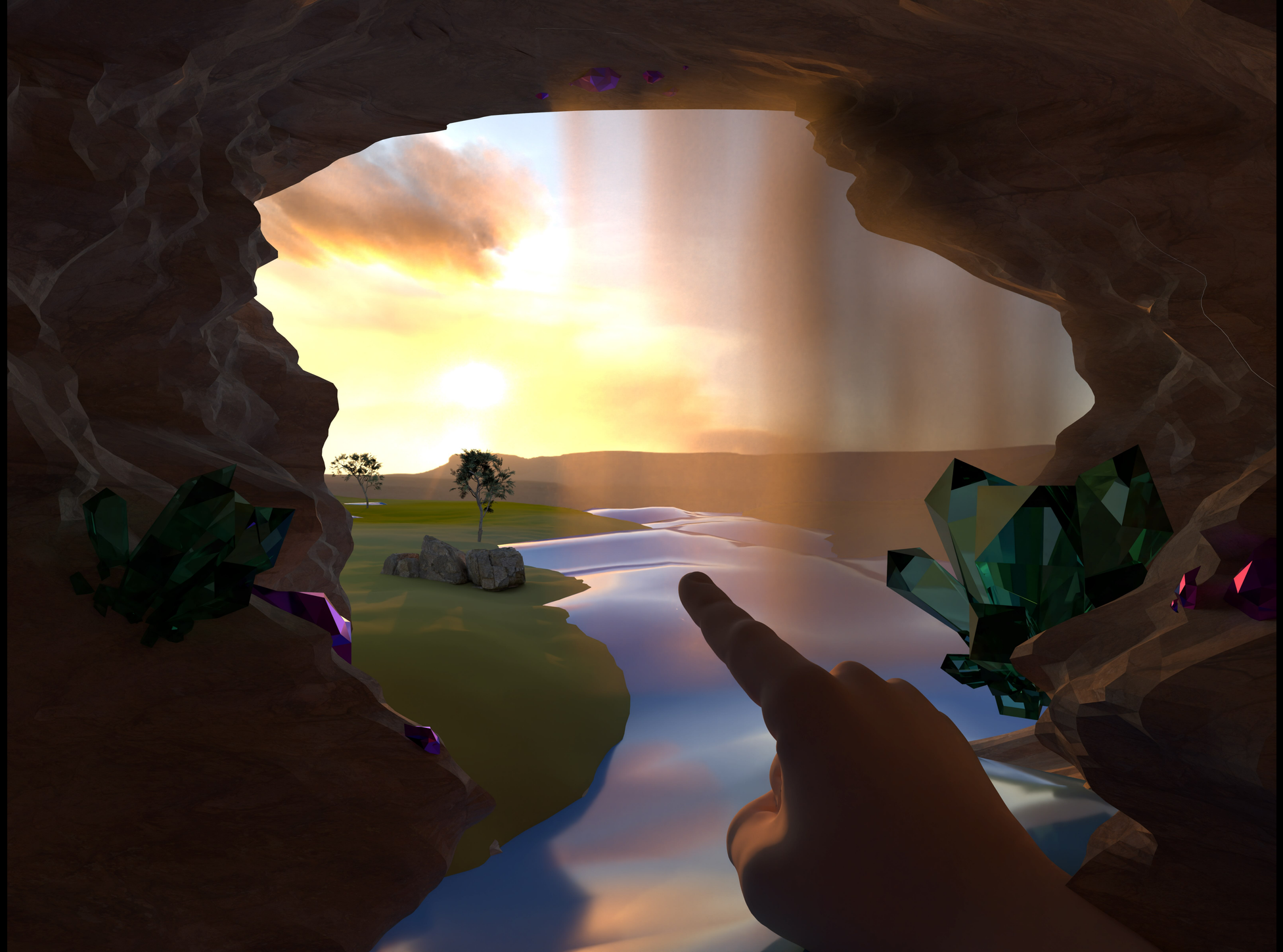
Original



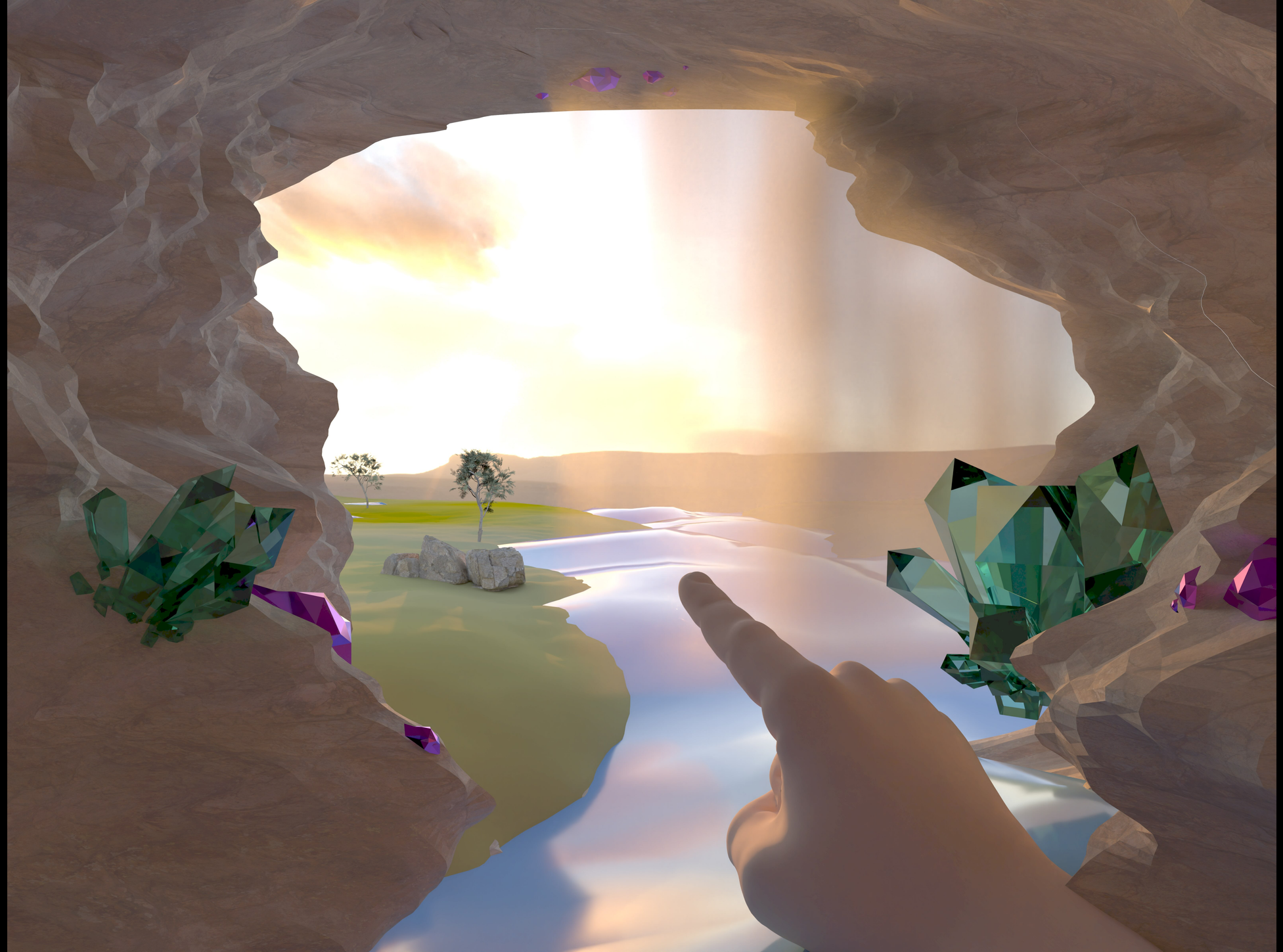
Denoised

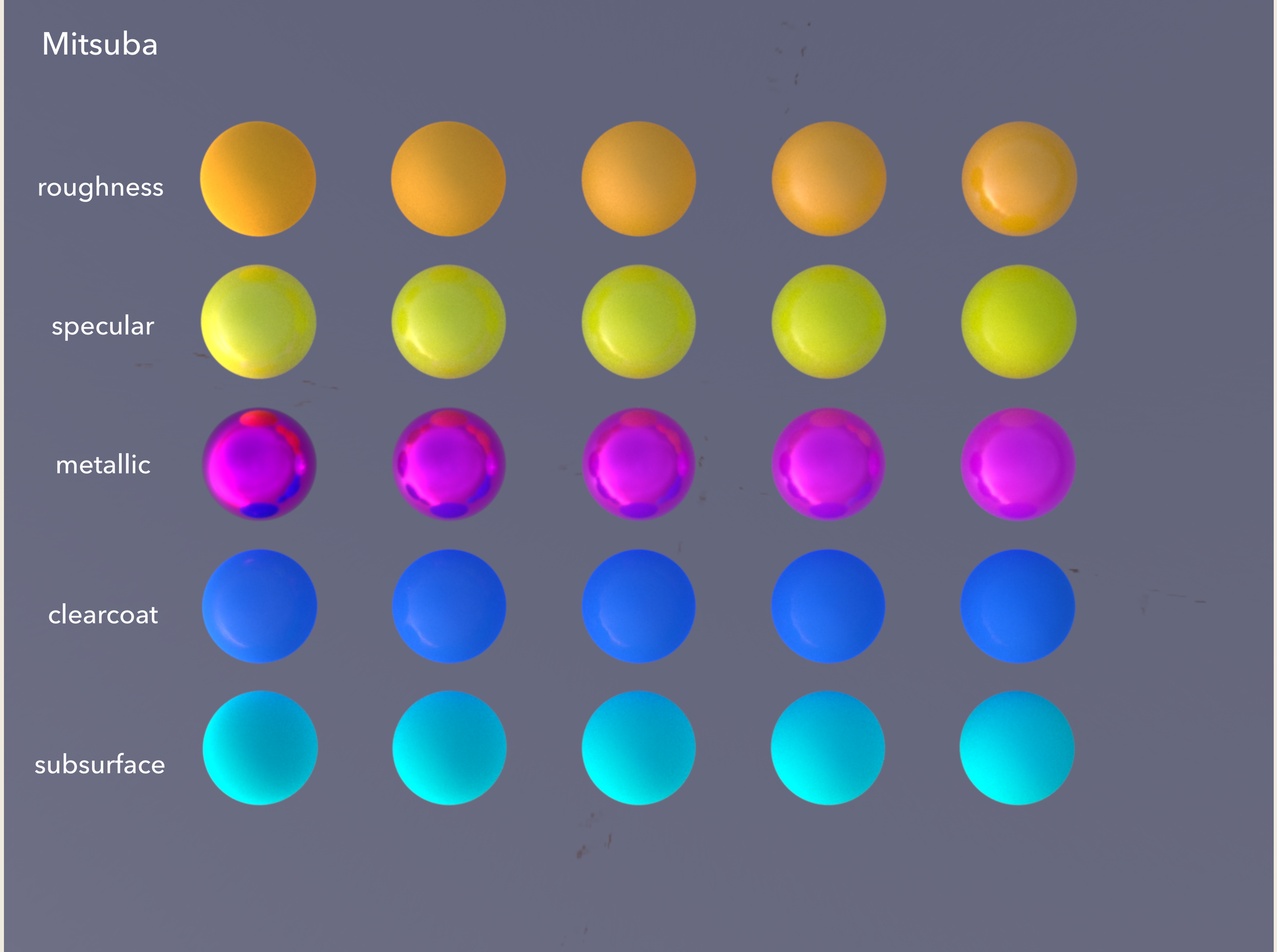


Denoised



Denoised




Mitsuba
roughness
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